



GSFC LNEXT CMO
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Landsat 10 Mission Synopsis

Update March 2026

L10-MGMT-DESC-0028, Revision March 2026 (DRFP
Version)
Landsat 10, Code 426

L10 Reviewed – Not Subject to Export Control

Previous releases: Feb. and July 2023 with MOC RFI,
Feb. 2024 with MOC studies
Jan. 2025 with S/C studies and industry days



- The Landsat 10 (L10) mission synopsis is a notional overview of the mission architecture and segment functions intended to provide high-level context for the objectives of the Landsat 10 mission.
- **There are no requirements contained in this document**, and it should be considered as reference material only.
- Information on requirements definition and compliance is provided in the L10 Spacecraft Statement of Work (SOW) and associated requirements documents.

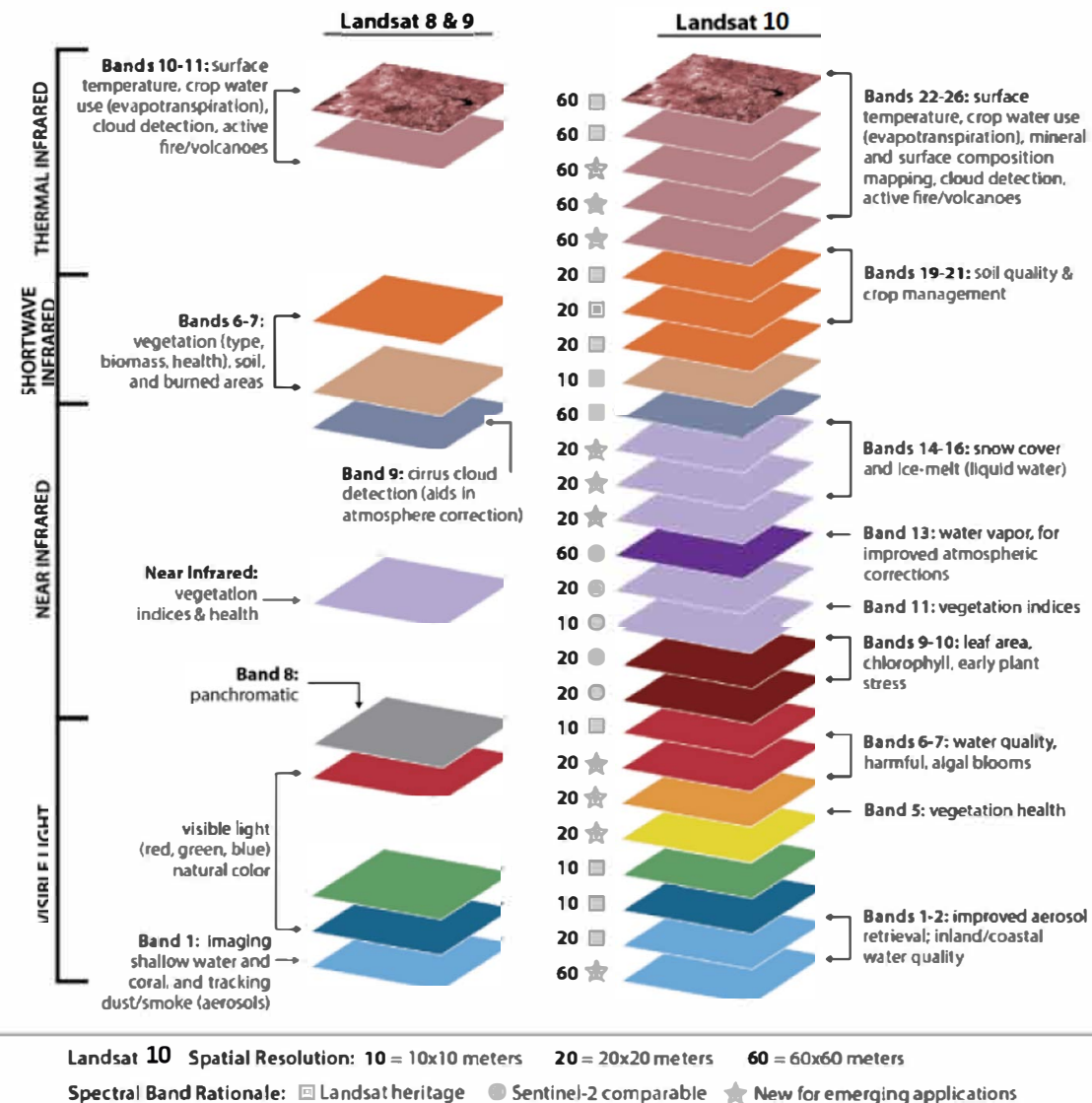
- Motivation for Landsat 10
- Landsat 10 Mission Background and Overview
- NASA/USGS Partnership
- Architecture & Mission Summary
- Space Segment Capabilities and Highlights
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- Ground Segment Architecture
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- Landsat Science and Products

- ***Landsat 10 will continue the 50-year Landsat data record***
 - Landsat is the ***most widely used land remote sensing data source within Federal civil agencies***
 - OSTP-led assessment ranked Landsat ***2nd highest in societal benefit*** (GPS was #1)
 - ***Landsat has dominated science literature since inception***, despite other open access EO sources such as MODIS
 - ***Commercial data providers need Landsat's rigorous calibration*** to build and improve their products
- ***Provides enhancements and new capabilities for the next generation of Landsat users***
 - ***Improved spatial resolution*** for agricultural monitoring, ecological monitoring, urban studies, water resources management and other applications.
 - ***Synergy with European Sentinel-2*** bands allowing easier merging of information products.
 - ***Improved spectral resolution*** to support new and evolving applications, including surface water quality, cryospheric science, geology, and agricultural applications including crop management and water consumption.
- ***Continues substantial economic benefits to the U.S. economy (>\$25 billion/year)***
 - Landsat data are used to understand our dynamic planet and forms the backbone of the US Sustainable Land Imaging (SLI) Program, returning an estimated ***\$2 billion/year in benefits to users***.
 - Information from ***Landsat contributes to day-to-day decisions on land, water, and resource use*** that protect life and property; safeguard the environment; advance science, technology and education; support climate resiliency; and grow the U.S. economy.

- The Landsat 10 (L10) mission is a joint mission being formulated, implemented, and operated by the National Aeronautics and Space Administration (NASA) and the Department of Interior (DOI) United States (U.S.) Geological Survey (USGS)
- L10 is a remote sensing satellite mission that will continue to acquire images of the Earth's land surfaces, thereby continuing uninterrupted data to assist land managers and policy makers make informed decisions about natural resources and the environment
- This mission continues the more than five decades of global data collection and distribution provided by the Landsat series of satellites
- The mission is a component of the Sustainable Land Imaging (SLI) Program conducted jointly by NASA and USGS of the DOI
- Major L10 mission objectives include:
 - Collect and archive moderate resolution, Visible to Shortwave Infrared (VSWIR) reflective and Thermal Infrared (TIR) emissive image data affording seasonal coverage of the global landmass for a continuous period of not less than five (5) years.
 - Ensure that Landsat 10 data are sufficiently consistent with data from the earlier Landsat missions in terms of relative acquisition geometry, calibration, coverage characteristics, spectral characteristics, output product quality, and data availability; all of which permit scientific studies, management of land cover and land use change, along with associated inland and coastal water resources, over multi-decadal periods.
 - Ensure Landsat 10 is responsive to documented emerging user needs and applications as characterized by periodic assessment, currently the User Needs for the SLI Program July 2018, Release 2.0, and identified by the operational requirements for acquisition, processing, archiving, and distribution of Earth's image data to the United States Government and other users.
 - Distribute Landsat 10 data products to the public freely on a nondiscriminatory basis.

- Landsat 10 is the next-generation NASA/USGS Earth-observing mission
- Launch Readiness Date: NLT December 2031
- Improve all spectral and spatial capabilities, while maintaining historical Landsat data continuity
- Landsat 10 will use scenes defined by new WRS-3
 - Landsat 7, 8, and 9 scenes defined by WRS-2

Parameter	Value
Mission Category	Category 2, Class C
Mission Life	5-years
Altitude	653 km Sun-synchronous
Inclination	~98 degrees
Mean Local Time Descending Node	10:10 AM +/- 5-min
Obs. Repeating Ground Track	18-day
Swath Width	164 km
Half Angle FOV	7.2 degrees



Landsat 10 NASA / USGS Partnership

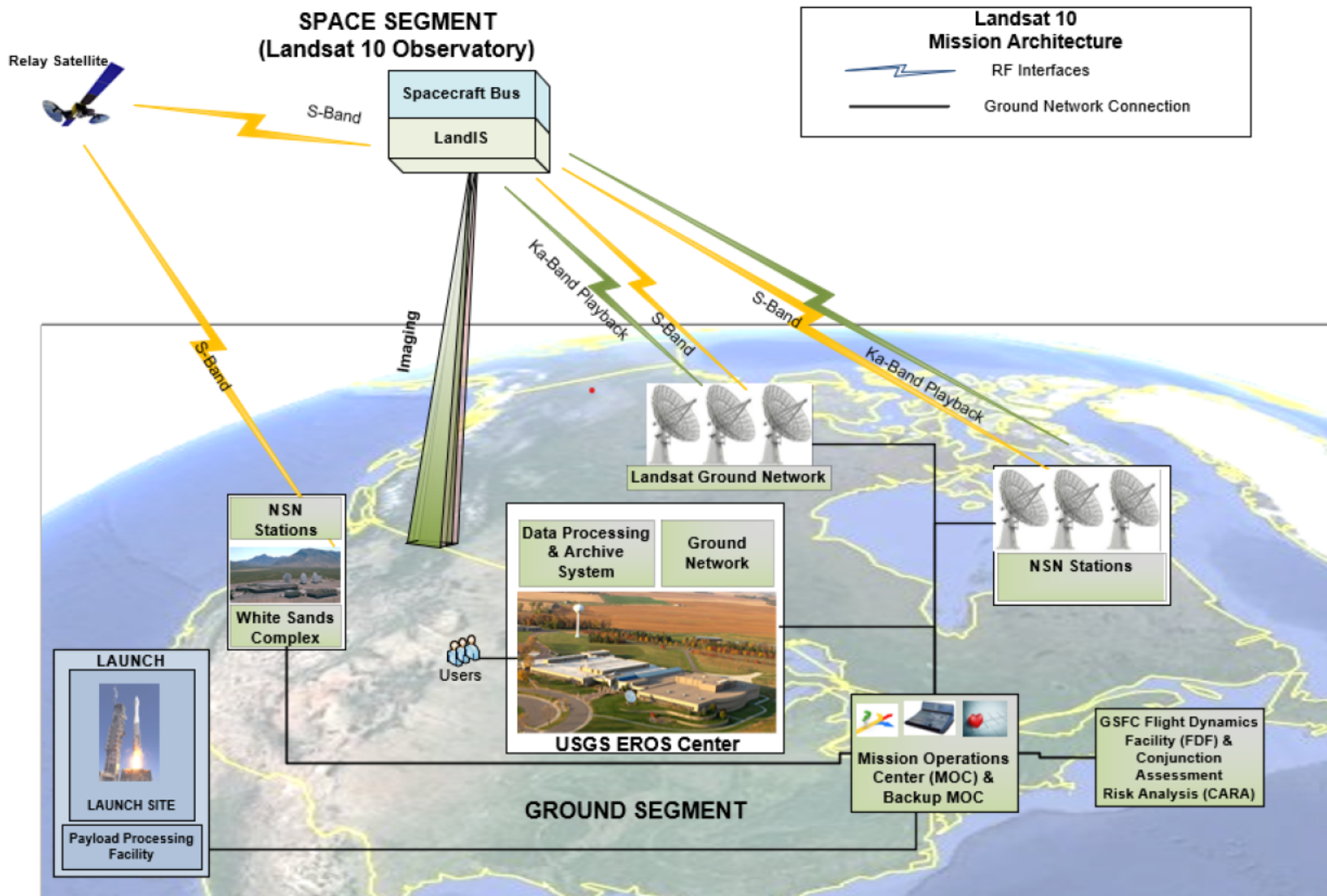


- NASA and USGS partner to provide independently funded joint-agency partnership
 - Handover of mission to USGS for continued operations following completion of on-orbit commissioning
- NASA is responsible for space segment (instrument(s), Spacecraft), launch and on-orbit checkout
 - Includes engineering support for mission operation readiness and support for on-orbit anomalies
- USGS will build and deliver the ground segment, operate the observatory after commissioning phase and provide data processing and distribution
 - Systems engineering support during development
 - Mission Operations Center (and backup)
 - Ground Network
 - Data processing and Archive System (DPAS)
 - Provide MOC capabilities
 - Provide Flight Operation Team



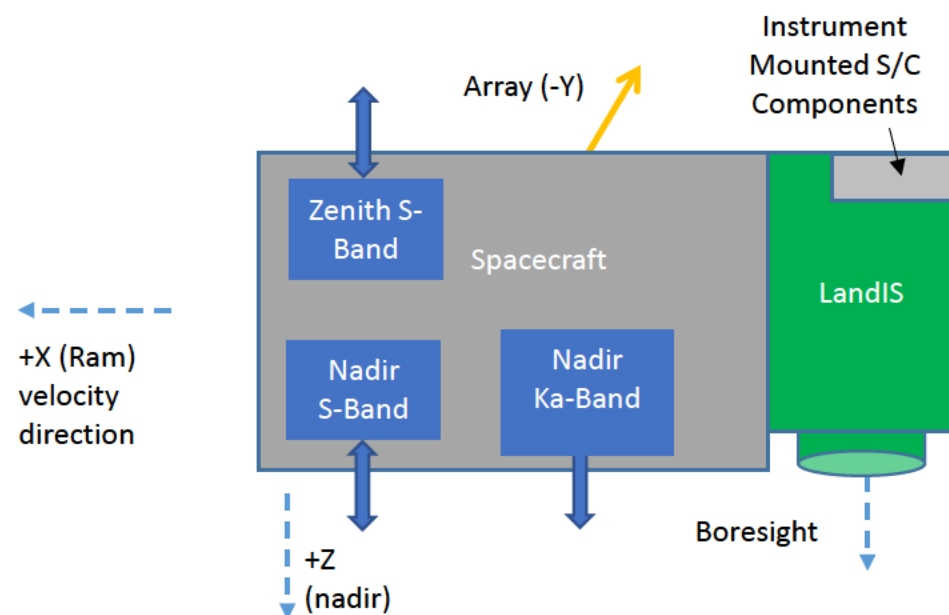
Same Roles and Responsibilities as Landsat 9

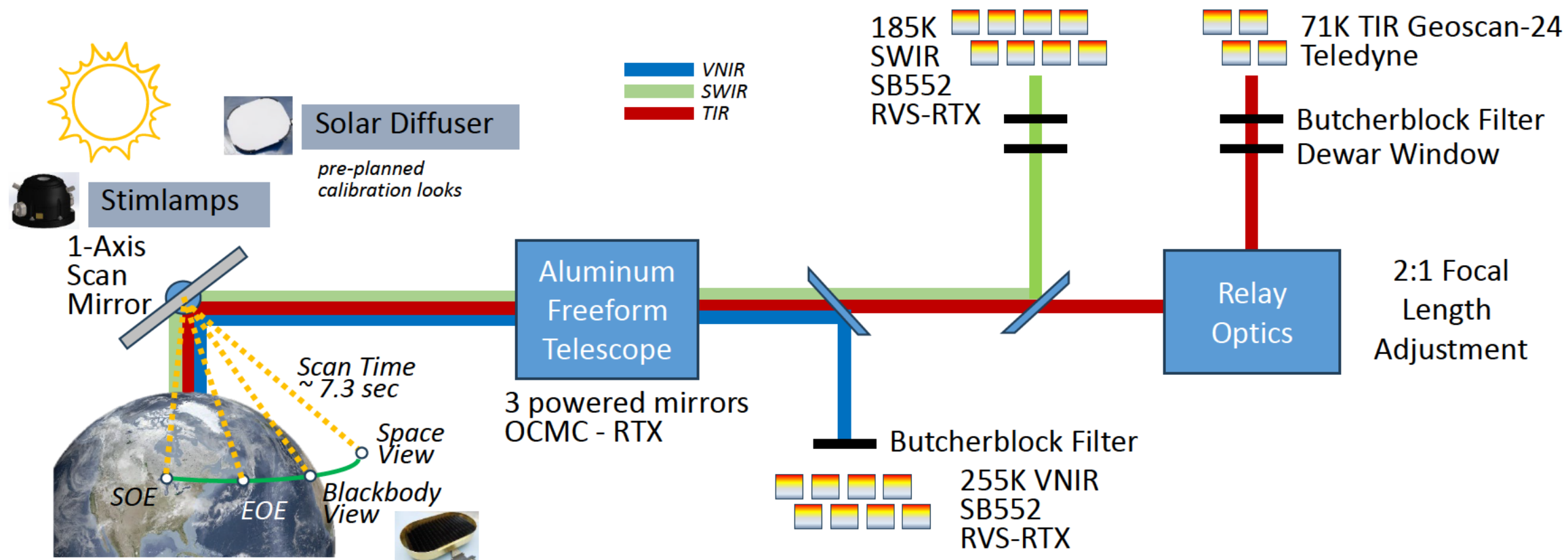
Mission Operational Concept Architecture



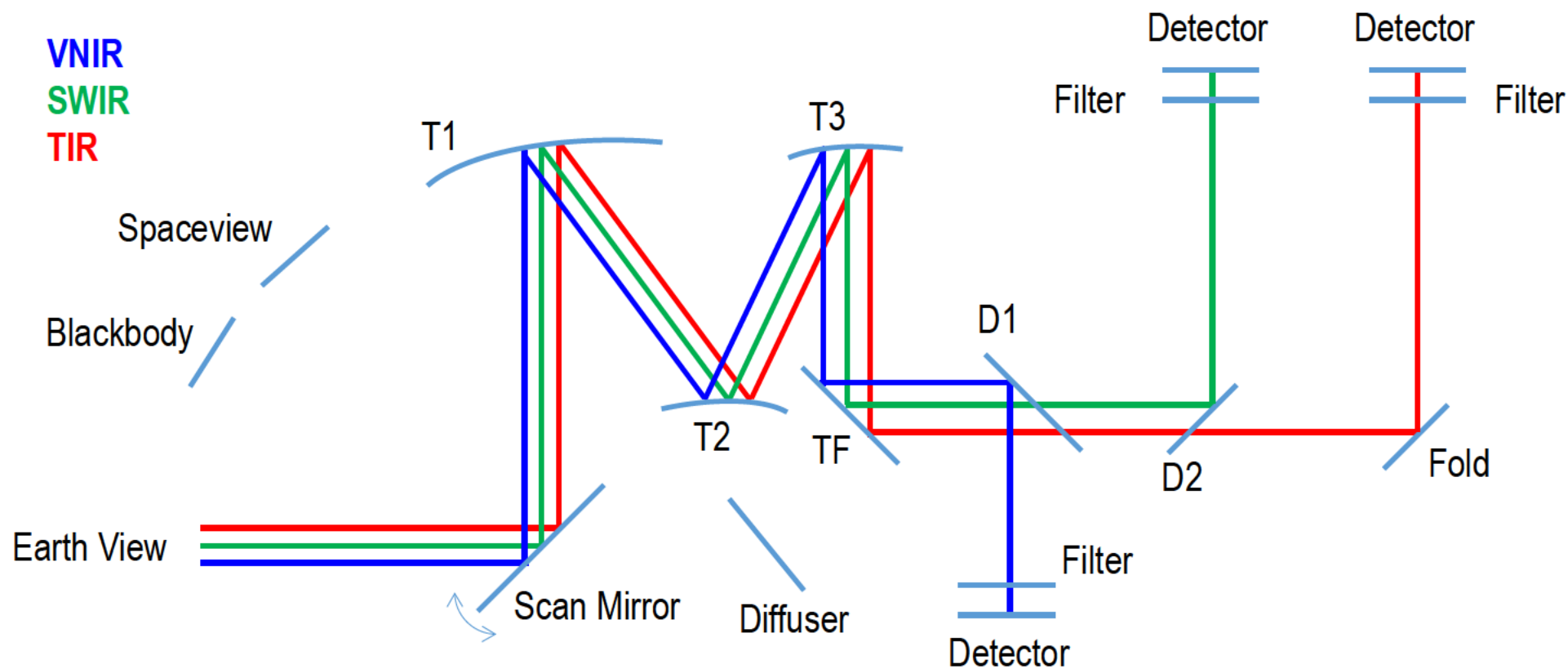
Landsat 10 Common Mission Architecture Diagram (L10-SYS-DESC-0022, Rev. B)

- The Landsat Instrument Suite (LandIS), a whisk-broom Earth-imaging scanner with twenty-six spectral bands and onboard calibration sources
- Autonomous imaging based on an on-board land database and on-board ephemeris, including all nominal calibrations to maintain image quality
- Attitude Control System that provides an array of sensors and actuators to provide precise pointing knowledge to ensure image quality and provide necessary Cal/Val special maneuvers (lunar calibrations, off pointing)
- Propulsion system supports orbit-control capabilities including drag makeup, inclination, orbital debris avoidance, and repositioning maneuvers, as well as controlled deorbit
 - 10 years (minimum) of fuel
- Instrument will be mounted on anti-Ram
- Ka-Band downlink of mission data (imaging sensor data, along with instrument and spacecraft ancillary data)
- S-Band downlink of housekeeping data (both real-time telemetry and stored State of Health (SOH) concurrently)
- S-Band decryption of uplinked commands
- S-Band communications providing near full spherical omni-directional coverage

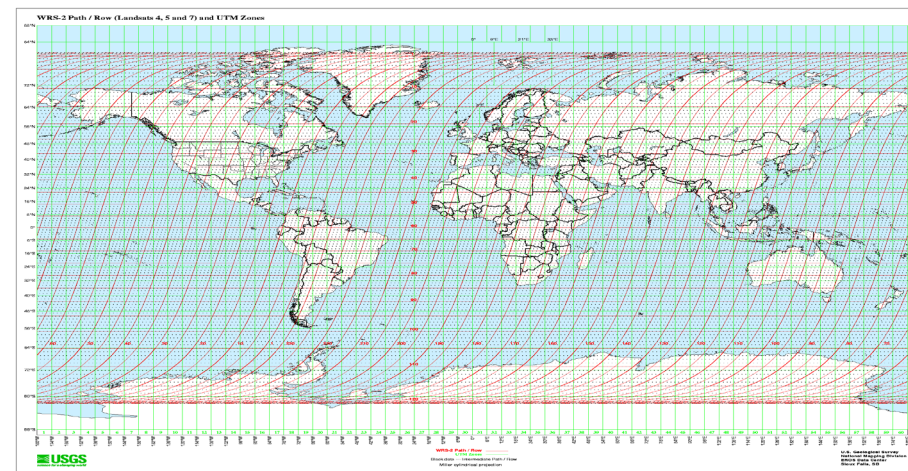




Single instrument whiskbroom design

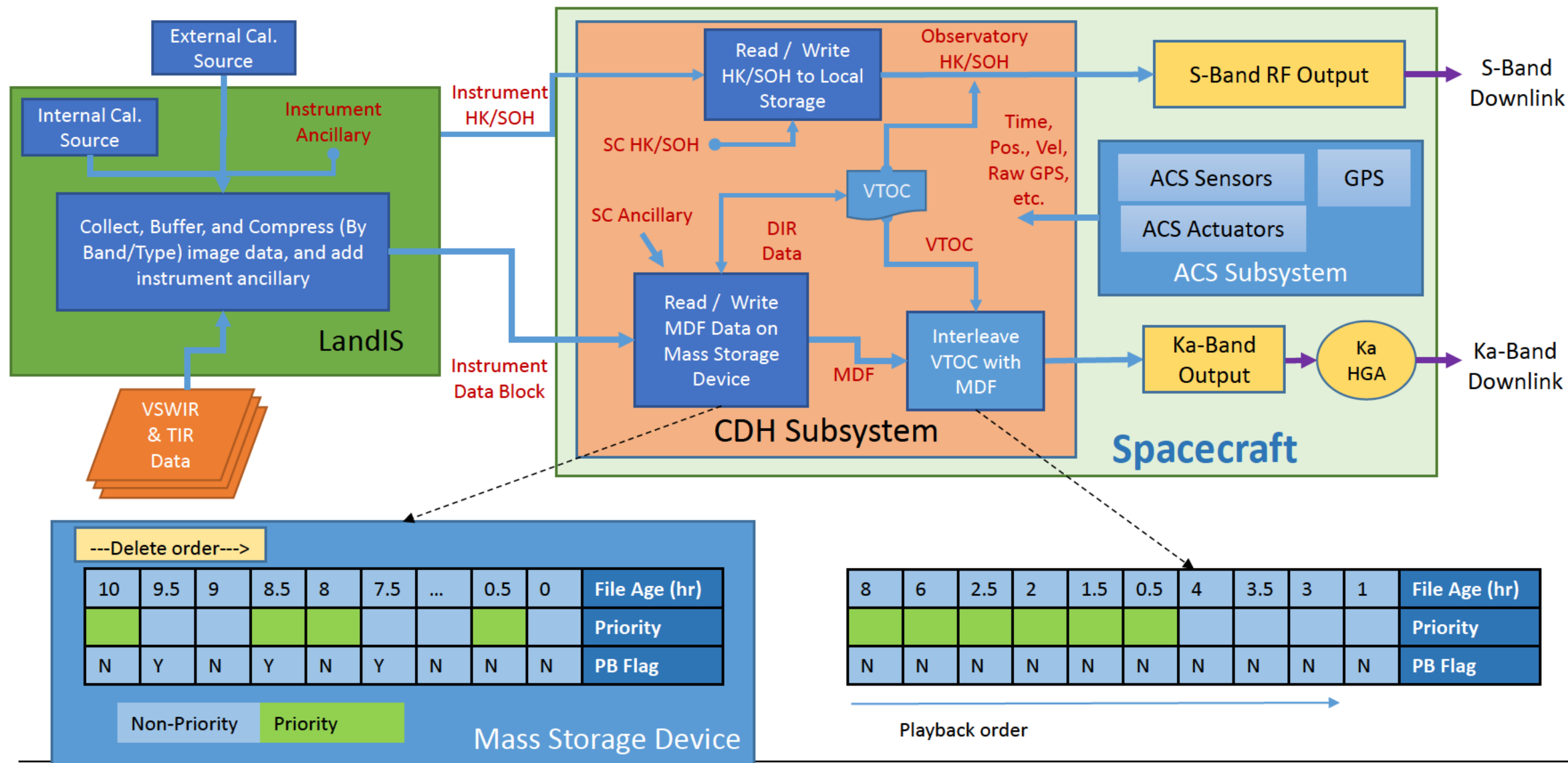


- L10 mission will utilize a Long-Term Acquisition Plan (LTAP) Collection Request (LCR) that will be stored on board
 - The LCR uses the new World Reference System-3 (WRS-3) grid system to identify various types of L10 intervals/scenes such as land, coastal, or open water regions
 - The LCR will nominally contain all sunlit land and a select number of nighttime images
- The observatory will acquire all land scenes as defined in the LCR and all nominal calibrations
 - Observatory is responsible for enforcing the solar elevation constraint (>5 degrees) for sunlit scenes
- Other special imaging requests, such as imaging of natural disasters or unique science research campaigns, and calibration collections are coordinated via a Special Collection Request (SCR)
- The observatory will be able to collect off-nadir image data up to ~ 15 degrees on either side of the satellite ground track, referred to as “one path over” or “off-nadir” imaging
- The observatory will also be able to perform off-nadir calibrations, which include lunar imaging, solar calibrations, and other instrument or attitude sensor calibrations
- Daily collection rates vary by how much land is overflown— daily average data volume expected to be approx. 12Tb



WRS-3

Observatory Data Functional Block Diagrams



- A mission data file contains an instrument data block and 30 seconds of spacecraft ancillary data, with a maximum size of 3.5GB
- A nominal science instrument data block is one, ~7.3 second scan and instrument ancillary data
 - Approx. 5 seconds of Earth imaging (View Zone 1), compressed by band and formatted into 52 data cubes
 - Uncompressed calibration data: Blackbody (View Zone 2), Space-view (View Zone 3), 52 data cubes each
- For diagnostic purposes, an instrument data block can also consist of diagnostic information and uncompressed data, as requested
- 30s of spacecraft ancillary is at least 12s pre-image and at least 10s post-image
 - 8s (minimum) of ancillary is required for science processing and to initiate the position model for an interval
 - 30s, with pre- and post- image ancillary, enables the ground system continue processing through a lost file during an interval
- LandIS will perform compression, and will utilize the CCSDS 123.0-B-2 compression engine
- Compression is tunable by band

- Each downlinked Mission Data file will be interleaved with an updated Volume Table of Contents (VTOC) with flags for Playback Status ("Not Played Back" or "Played Back"), priority, and protection so the Ground System can determine the latest status of every file on the Mass Storage Device (MSD)
 - The VTOC is updated whenever a file is created, downlinked, or its state is changed by command
 - The Ground can select mission data to be re-downlinked by commanding a change to the "not played back" flag per file
- CCSDS Virtual Channels (VCs) will be used to separate priority and routine (non-priority) mission data files
 - VC for priority data provides an approximate analogue to the Landsat 8 and 9 real time downlink capability for selected areas
 - However, there is no capability to transmit priority data only
 - Allows for simplified identification of priority data on the ground for expedited delivery and processing
- Mission data files transmitted to the ground are unique and will not be re-transmitted unless by ground command
- Ground schedules communications link configurations and contact and downlink start/stop times

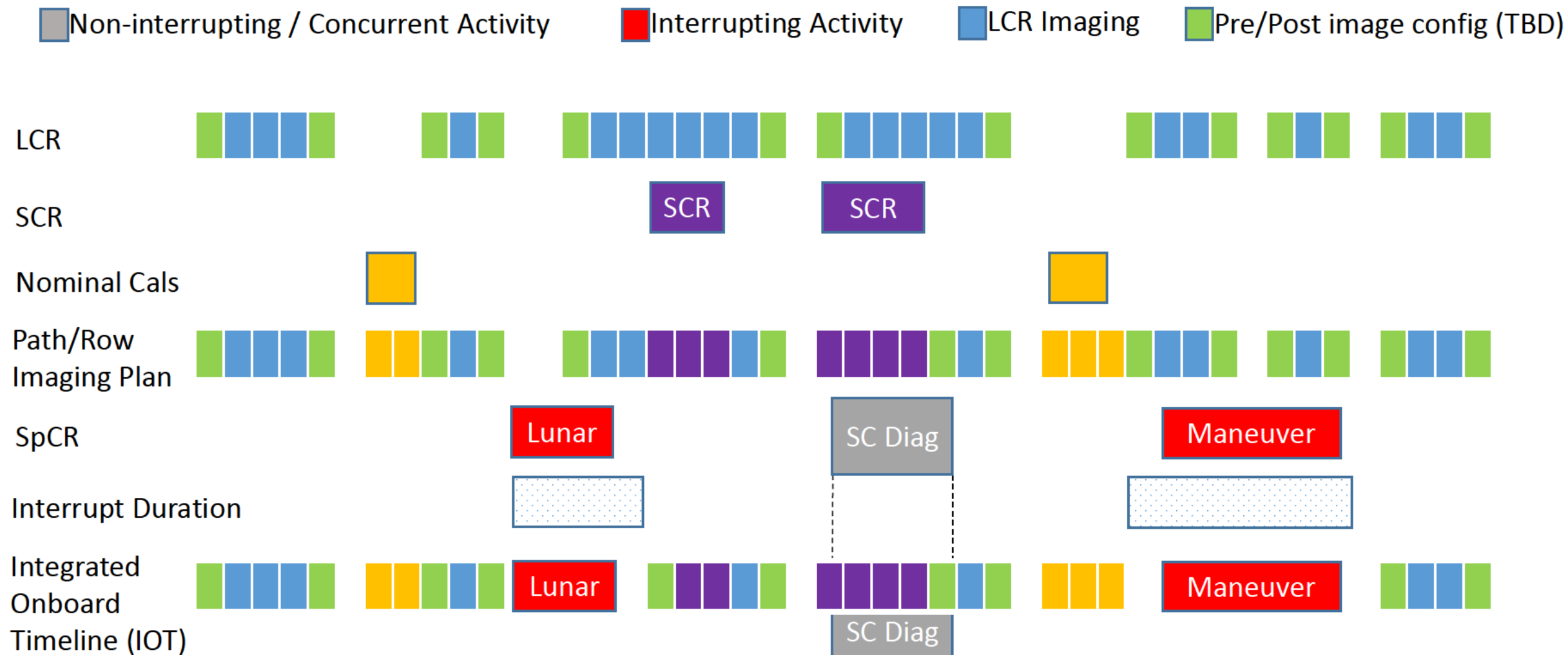
- The spacecraft mass storage system will be sized to ensure 8Tb of mission data is available for downlink
 - All files, except those marked as “protected”, can be overwritten whether they have been played back or not.
 - Overwriting, or deleting in advance of writing, will be done based on time (age of file) and storage need (only overwrite when memory is needed for new data recording), and will not be based on priority or playback status.
 - The spacecraft will overwrite oldest data first, to enable the maximum time for downlink and retransmission.
- Ground has ability to mark a file as "protected", which will only be used in contingency scenarios, but will preserve the file from being overwritten. This will only be used for diagnostics or anomaly investigations, and not routinely.
- Mission data on the observatory will be managed by autonomous capabilities of the on-board flight software, as well as directed commands from the Ground as needed to handle non-nominal operational scenarios.
 - Mission data files are downlinked by Priority then by Time – priority first, then non-priority, in age order
 - Any change in a file status (new file recorded, played back file marked not played back, file played back, etc.) will trigger a re-order of the downlink order – this ensures Priority files recorded or changed during a contact are downlinked “next”

- S-Band and Ka-Band contacts will be highly automated to support lights out operations
 - Contacts can be S-Band only, Ka-Band only, or both
 - All contacts are planned and scheduled by the ground, and the confirmed contact schedule is loaded to the observatory on a periodic basis (approx. weekly, TBD)
 - The ground can replace the onboard schedule, as well as add or delete individual contacts, as needed

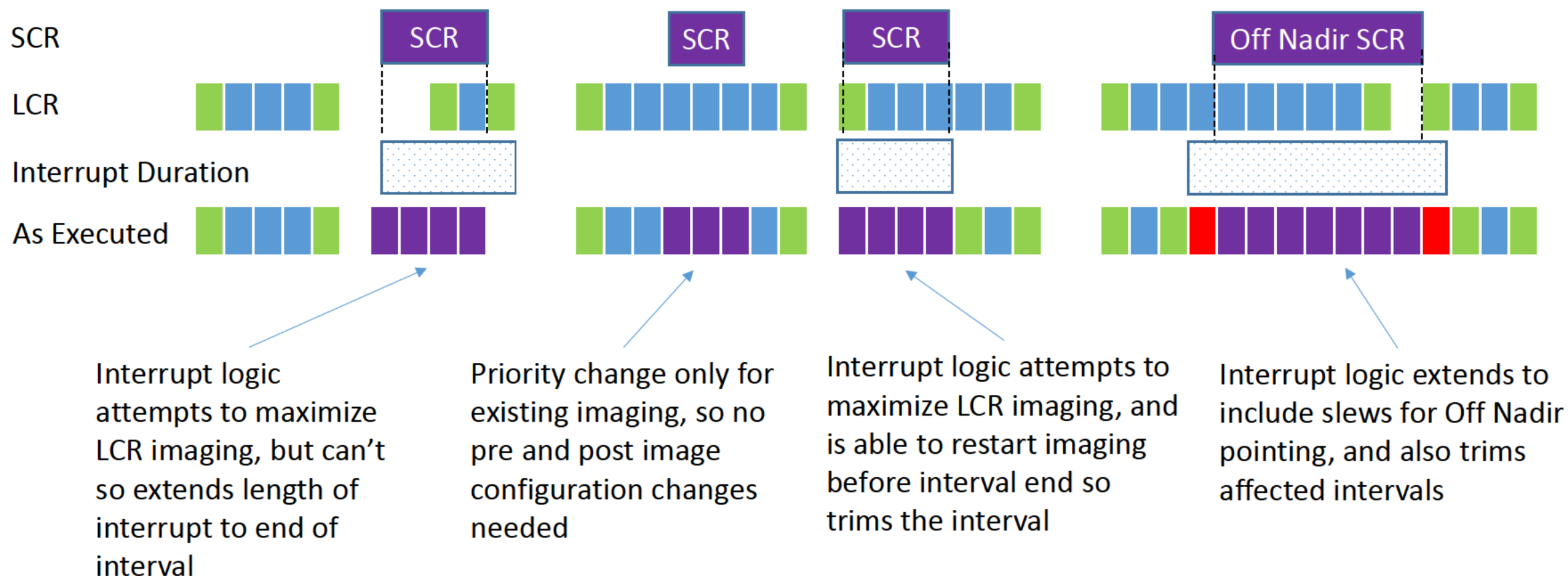


Ground System	Spacecraft (per observatory)
• Plans and schedules ground resources • Builds and Loads contact schedules to observatory • Autonomously manages ground resources • Receives real-time SOH and stored SOH via S-Band • Receives Mission Data via Ka-Band and manages retransmit needs • Sends retransmit commands via S-Band to S/C	• Converts contact schedule to commands • Manages antenna pointing based on contact schedule and knowledge of Ground Stations • Executes pre/during/post contact commanding • Executes ground received commands during S-Band contacts

- Fully autonomous nominal WRS-3 Scene (Path/Row) imaging
 - Built from the LCR, which will be updated by the ground infrequently (approx. 1-2 times per year)
 - Special collection requests (SCRs) replace the baseline LCR scenes, until removed by the ground
 - Includes capability to do one path over (either direction) off-nadir acquisitions
 - LCR/SCR based imaging can continue indefinitely, without ground interaction, unless halted or directed to change by the ground or onboard Fault Management
 - Autonomous imaging includes all pre and post imaging configuration, all nominal path/row based calibrations, and requested night imaging
- Special Command Requests (SpCRs) are ground-directed (not autonomously generated by the Spacecraft), individual time based requests, for special activities outside of nominal scene based imaging
 - Includes spacecraft or instrument activities such as propulsive maneuvers, diagnostic activities, or maintenance activities
 - Includes infrequent calibration activities such as lunar calibrations or bi-annual collections
 - SpCRs can be interrupting or non-interrupting (able to be concurrent with imaging)
- Important Integrated Onboard Timeline (IOT) generation highlights:
 - SpCRs take precedence over any SCR or LCR based activity.
 - SpCRs are based on time, while SCRs and the LCR are based on Path/Row.
 - Interruption logic expands interrupt “cutout” duration as necessary to ensure the safety of the observatory at all times (not leaving instrument or S/C in an unknown or unsafe state), and maximizes imaging whenever possible.
 - If an SpCR is removed (ie. a collision avoidance burn is deleted), the LCR/SCR-based imaging fills back in (resumes, repopulates, etc.) as if it was not interrupted.



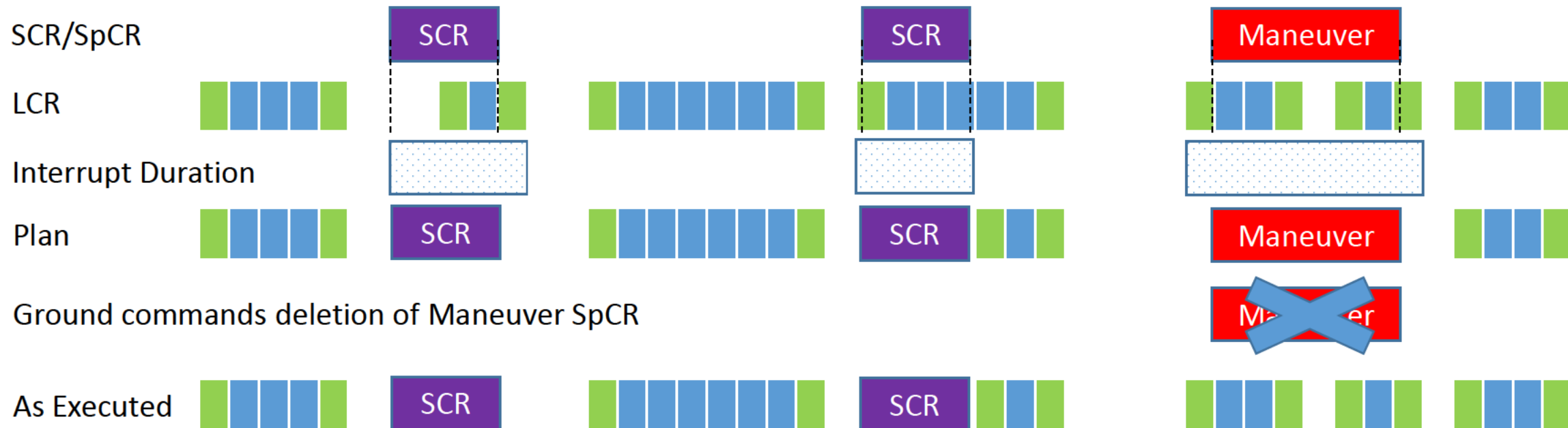
Imaging and Special Request Logic and Deconfliction Examples

 Slew  Imaging  Pre/Post image config (TBD)**SCRs and SpCRs take precedence over LCR imaging, "interrupting" the nominal LCR timeline**

Imaging and Special Command Request Logic – SpCR Removal

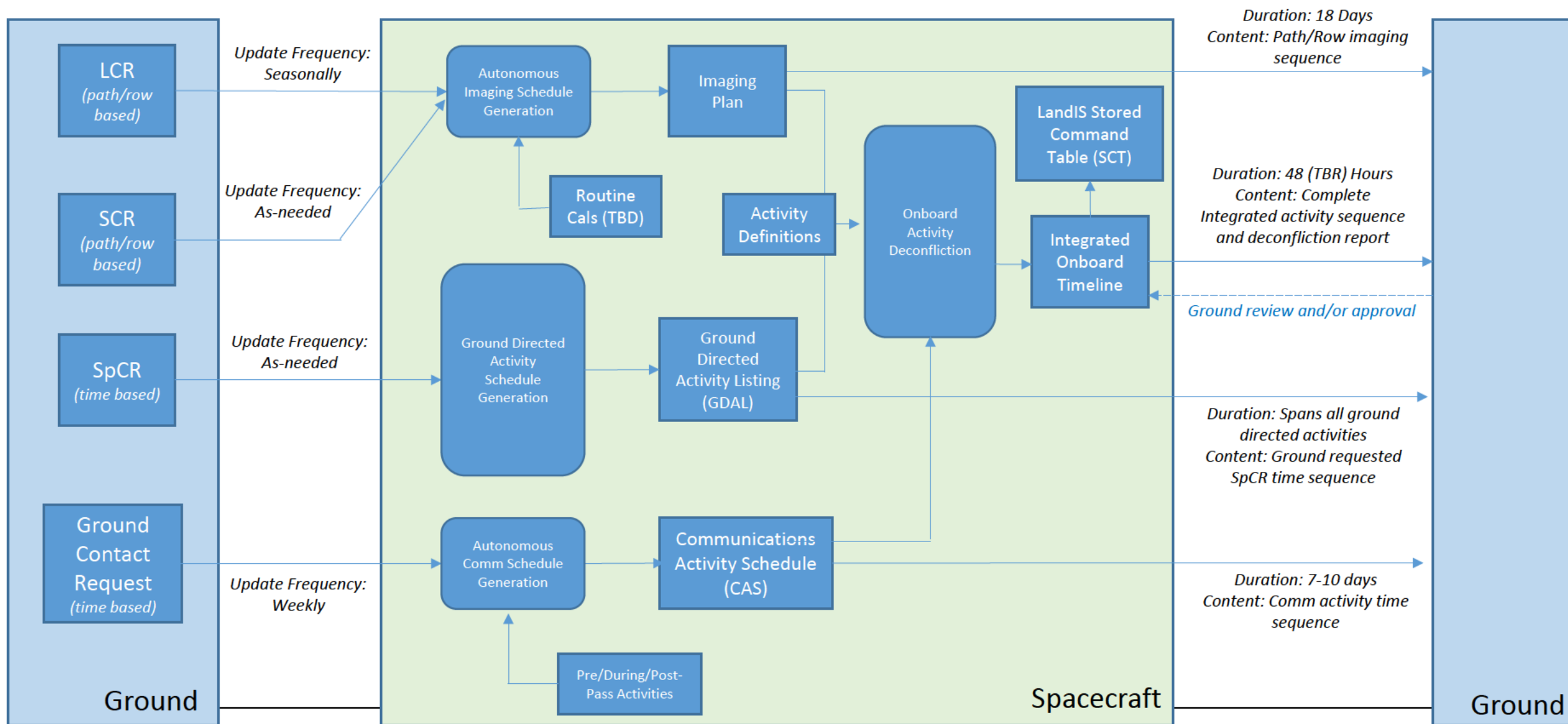
Imaging

Pre/Post image config (TBD)

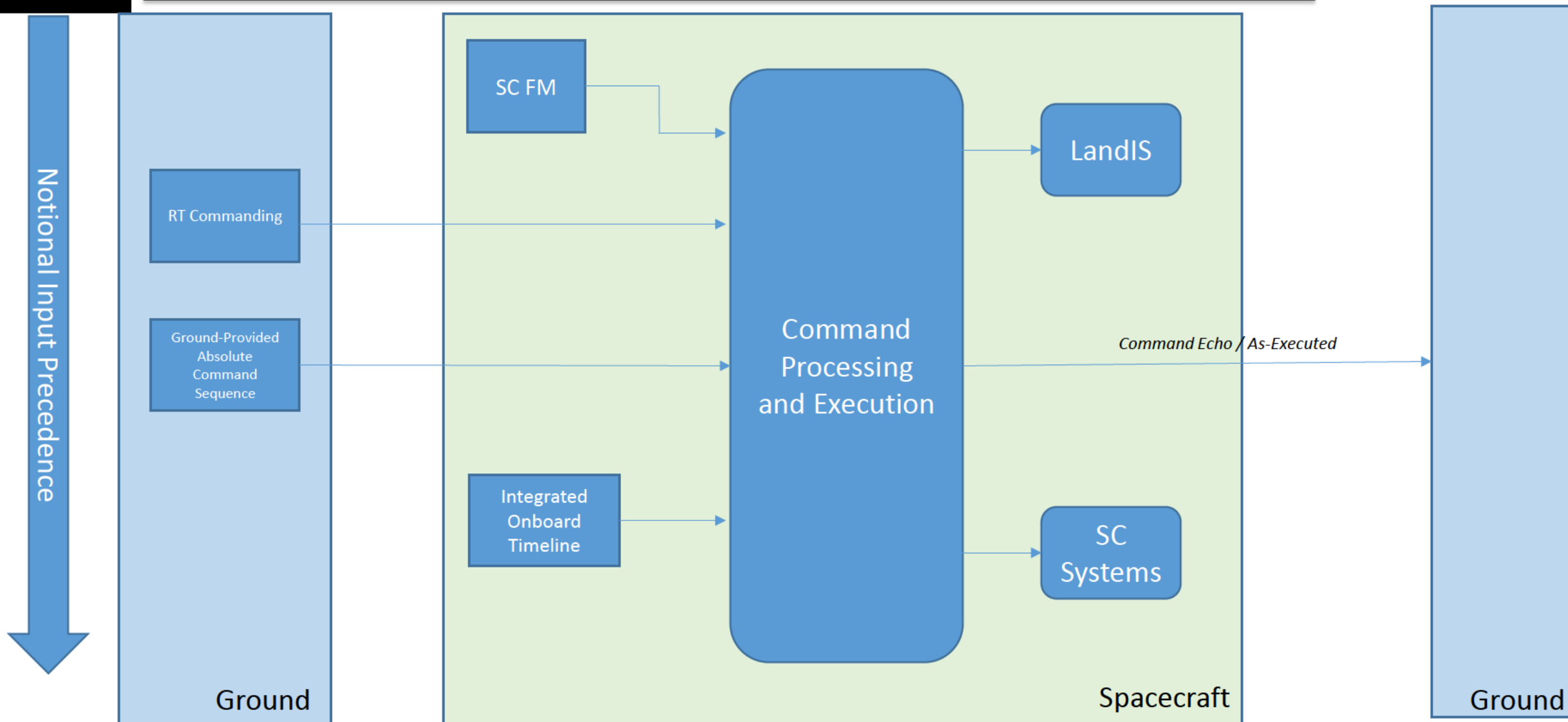


S/C logic attempts to maximize LCR imaging, and fills back in LCR imaging once Maneuver is deleted by ground command

Once SCRs/SpCRs are deleted the spacecraft will revert to LCR imaging



L10 Command Input Paths (Notional)



SC FM will always supersede any onboard loads to ensure the safety of the observatory

L10 Ground System Input to Spacecraft

- The ground provides a limited set of inputs to the spacecraft, from which the spacecraft will generate onboard the integrated timeline to support imaging, special activities, and comm links. These ground provided inputs are the following:
 - Ground Contact Request (GCR) - Communications Schedule*
 - Uploaded ~weekly
 - *Process for generation of the onboard communications schedule is separate from the imaging sequence and comm activities may be executed out of a separate onboard load sequence.*
 - LTAP Collection Request (LCR)
 - Listing of all nominal scenes - **defined by path/row** - to be acquired by the spacecraft, as defined in the LTAP
 - Fairly static, expected to be updated infrequently
 - Special Collection Request (SCR)
 - Individual requests for acquisitions outside of the baseline LCR. **Defined by path/row.**
 - Requested regularly on an as-needed basis
 - Special Command Request (SpCR)
 - Individual requests for singular operational activities (e.g. maneuvers). **Defined by execution time.**
 - Requested regularly on an as-needed basis
- Key Assumptions
 - The ground will provide SCRs that do not conflict with each other
 - The spacecraft has responsibility for deconflicting the LCR with SCRs/SpCRs as requested from the ground – the ground does not resolve the SCRs/SpCRs with respect to the LCR prior to upload
 - Activity Definitions maintained onboard are used during conversion from activities to commands and deconfliction
 - The spacecraft has responsibility for autonomously generating the integrated onboard timeline and associated command sequences which support all the imaging and activity requests provided by the ground via the inputs listed above
 - The integrated onboard timeline generated by the spacecraft is provided to the ground for review
 - The spacecraft also provides intermediate schedule products to the ground for longer term planning activities

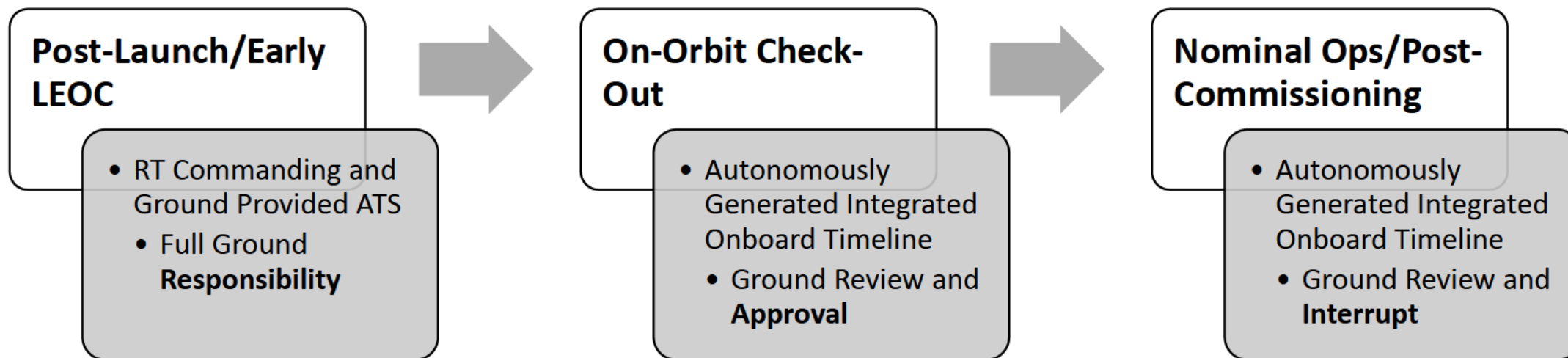
Ground provides a streamlined set of inputs which enable the spacecraft to generate the integrated timeline autonomously onboard

L10 Key Spacecraft Output to the Ground

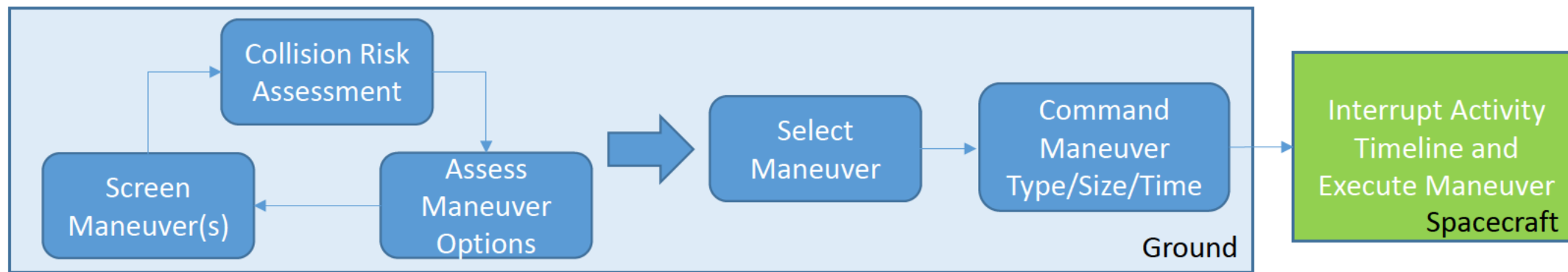
- The Spacecraft provides the following outputs to the ground
 - Imaging Plan
 - Duration: 18-days (Starting from last generation time)
 - Content: Path/row-based summary of all planned activities for full repeat cycle (includes LCR and SCRs)
 - Used for coarse long-term planning, and is based on the LCR and SCRs
 - Ground Directed Activity List (GDAL)
 - Duration: Span all planned SpCR activities
 - Content: Time-based plan of all special command request activities currently maintained onboard
 - Communications Activity Schedule (CAS)
 - Duration: nominally 7-10 days
 - Content: Time-based plan for all comm activities currently maintained onboard
 - Onboard Integrated Timeline (IOT) and deconfliction report
 - Duration: 48 hours
 - Content: Absolute command sequence the spacecraft is planning to execute on-orbit
 - Ground review and/or approval (LEOC)
 - Command Echo/As-Executed Timeline
 - Spacecraft will provide telemetry regarding the actuals executed on-orbit

Spacecraft provides a set of outputs of varying fidelity to ensure sufficient ground insight into on-orbit activities

- Operational IOT autonomy will ramp up during commissioning
 - Exact phasing of autonomy functions will depend on results of pre-launch testing, mission readiness planning, and spacecraft, instrument, and science team recommendations
- On-orbit demonstration of the core functionality of the succeeding phase is a gate to entering next phase
- All autonomous functions are configurable per requirements, and can be scaled back/overridden as desired throughout the mission
- The phases shown below can be thought of as Crawl/Walk/Run



- Nominal propulsive maneuvers will be highly optimized, to reduce the time between maneuver decision and execution, and to provide flexibility to the operations team
- The ground will request an interruption of the executing integrated activity timeline when a maneuver is needed
 - The spacecraft will fill back in any interrupted imaging if a maneuver sequence is deleted/removed
- Frequent collision threats are expected during the life of the mission, which drives the need for increased optimization and rapid turnaround – allowing the FOT to wait as long as possible before committing to a burn
 - The ground system will be able to plan and assess maneuver options for the FOT to select and command the spacecraft to execute
- Pre-loaded maneuver sequences will be available for selection by the FOT to execute a pre-determined range of in-track and retro maneuver options without the need to build and load a custom command sequence (TBD)
 - Infrequent or highly specific propulsive maneuvers can still be built, tested, and loaded from the ground, as needed



- The spacecraft will be capable of autonomously maintaining the safety of the spacecraft and instruments, in the case of a long ground outage or major onboard anomaly
 - This will include typical onboard fault management functions like power and thermal anomaly detection and responses (load shedding, maintaining power positive state, thermal stability, pointing keep out zone management, etc.)
- End of Life and Decommissioning
 - The observatory will carry 10+ years of consumables (fuel), with the goal of extended mission life and science operations
 - USGS, in collaboration with NASA, will make the determination to decommission
 - The Landsat 10 observatory will be disposed of in accordance with NASA and USGS policies, which includes controlled deorbit as the baseline

- Mission Integration is composed of the following:
 - Observatory Integration (LandIS to S/C Bus)
 - Environmental Test Campaign
 - Mission Readiness Campaign
 - Launch Vehicle Integration
- LandIS will be delivered to the S/C fully qualified
 - Instrument performance requirements will be verified at the instrument facility, and only instrument functional and limited performance testing will be repeated at the observatory level
- The S/C bus will follow commercial best practices up to the start of observatory integration
 - Will include, at a minimum, a baseline Comprehensive Performance Test (CPT) and verification of instrument interfaces
- Observatory integration and test will follow the 7123.1 I&T approach (ie. Plan, TRR, Consent to Break, post test review, test report, etc.)
 - Tests include vibration, shock, acoustic, EMI/EMC, self-compatibility, and thermal
 - Testing will include limited performance (i.e. end-to-end, day in the life)
- RF Suitcase testing (wideband only) will be performed at all LGN Ground Stations and select NSN stations
- Independent data validation will be performed by USGS throughout I&T and will include end-to-end data chain validation with the complete observatory and ground system (Photon to L1 products at EROS)

- Mission Readiness is operations focused and validates the mission conops and mission requirements – it demonstrates the team, systems, products, and procedures are ready for a safe and successful launch and subsequent operations
- The mission readiness campaign follows Test Like You Fly best practices and is structured to successfully pass through the following key mission reviews (see NPR 7123.1 for more detail):
 - Mission Operations Review (MOR), Flight Operations Review (FOR), and Operations Readiness Review (ORR)
 - Flight Readiness Review (FRR) and Launch Readiness Review (LRR)
- Incremental approach concurrent with Observatory integration, testing, and launch preparations
 - Flexible schedule, coordinated across the project, agencies, and vendor teams, that adapts to system maturity
- Fully integrated team participation, including NASA (Engineering and Mission Integration), USGS (Ground, Science, and Operations), Spacecraft vendor, Instrument vendor, and Launch Services
 - NASA leads and coordinates mission and flight readiness activities and USGS leads ground readiness activities
 - USGS provides core flight operations and ground system engineering teams
- Core tests include Network Readiness Tests (NRTs), Ground Readiness Tests (GRTs), Spacecraft Operations Readiness Tests (SORTs), Mission Readiness Tests (MRTs), and Mission Simulations
- Note: Observatory requirements verification is performed independent of MRTs

Simulator Key Highlights and Capability Summary

Simulator	Key Highlights	Description and functional details
Command and Telemetry Simulation System (CTSS)	- Simple C&T functions - Deployable internationally	- Simple, low fidelity system for simulating realistic spacecraft command and telemetry (C&T) responses with basic scripting capability - Primarily used for ground station and ground system checkout activities - Deployable to international partner ground stations with limitable command and telemetry database functions (ie. remove all but a no-op and acknowledgement for ITAR/CUI protection)
SoftSim	100% software - Runs S/C FSW only - Has full environmental truth model	- Software based (emulated hardware) simulator of the Observatory - High fidelity Spacecraft (executable/patchable flight software (FSW), modeled subsystems, and environmental truth model, etc.) - Low fidelity LandIS (basic C&T only) capability - Primarily used for spacecraft and ground procedure development, ground software development, and ground readiness tests
Observatory Operational Simulator (OOpS)	- Runs S/C and LandIS FSW - Has full environmental truth model - Workhorse simulator for ground, mission readiness, and operations teams	- High fidelity observatory simulators (integrated LandIS Operational Sim (LOpS) with S/C Operational Sim (SOps)) with flight like timing for all mission data elements - Flight-like command and telemetry response, redundancy, subsystem models, timing, compression, and science data generation/flow - Executes S/C and LandIS FSW, with ability to use and update as in flight - Primary resource for operational process and procedure development, Mission Conops validation, operations testing, and Mission Readiness tests and simulations - Long term operational use: special activity testing, product updates, anomaly investigation, team training and proficiency, etc.

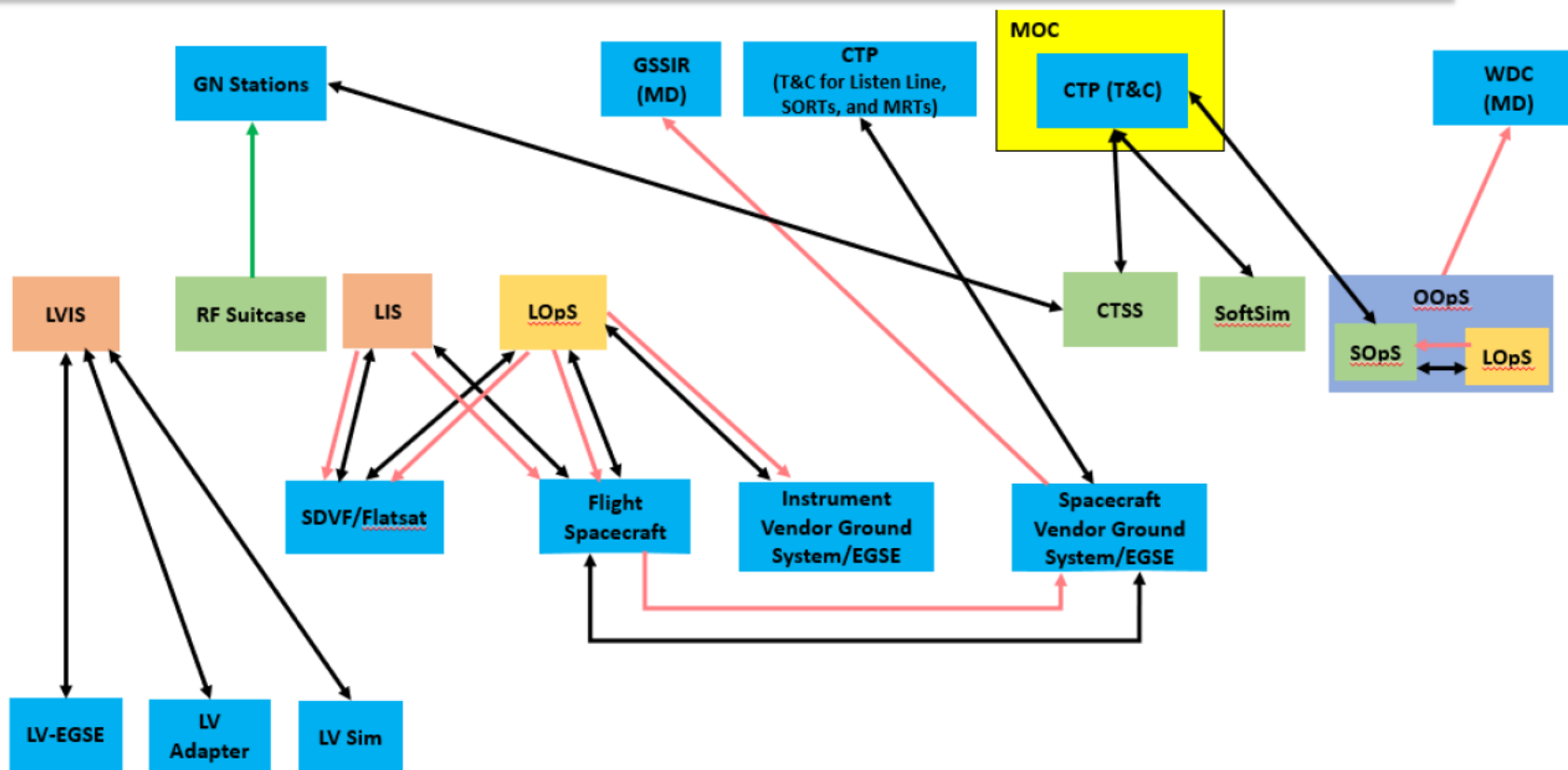
- The L10 simulators are intended to provide incrementally more capability as the flight and ground segment needs increase during the mission development, launch preparation, and operations phases
- L10 will also utilize interface simulators for interface verification:
 - LandIS Interface Simulator (LIS) – Early Instrument to S/C interface checkout
 - Launch Vehicle Interface Simulator (LVIS) – Observatory to Launch Vehicle interface
 - RF Suitcase – Realistic Ka-Band RF signal to Ground Station hardware (free-space and/or coax.)

Ground Segment Assets
(GS Req verifications and Ops Development)

Space Segment Simulators

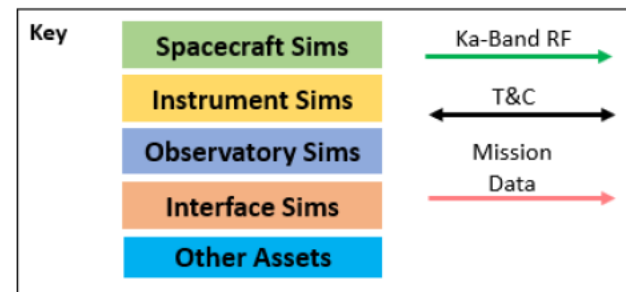
Space Segment Assets
(SC and Inst Req verifications)

Launch Segment Assets
(LV Req verifications)

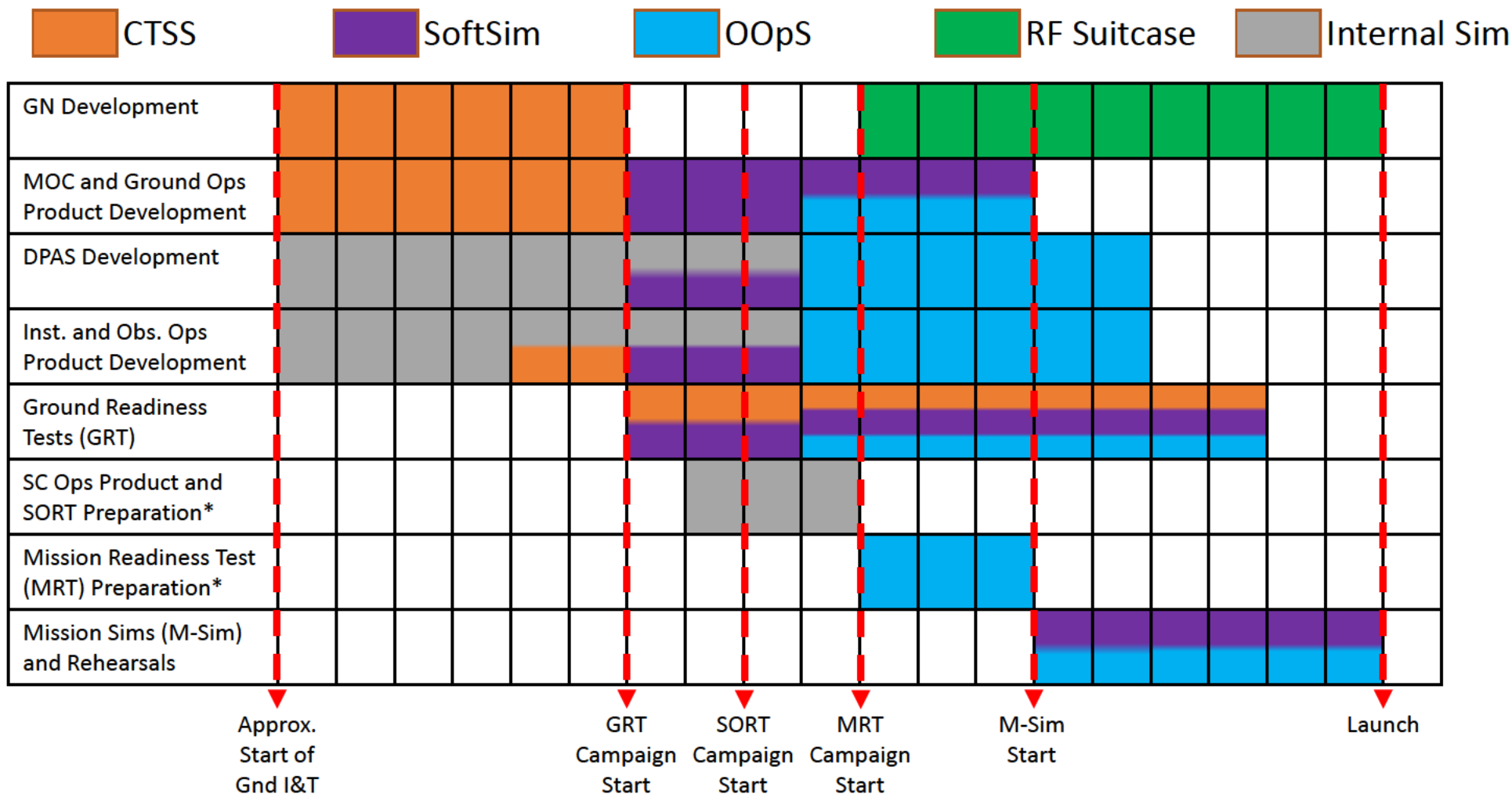


Notes:

1. All Sims can be used standalone
2. SDVF/Flatsat is notional – represents internal S/C capability that LIS and LOPs can be used with, if needed
3. The RF Suitcase is a Spacecraft Sim of only the Ka-Band System
4. CTPs are resident at the MOC or the GN Stations; A CTP will reside at the SC Vendor facility as the T&C interface to the MOC for testing and telemetry flow
5. The OOpS simulator is the combination of the LOPs and the SOpS. Interfaces of the combo are shown in the OOpS.



Notional Mission Readiness Simulator Uses and Timeline



Note: Timeline not to scale

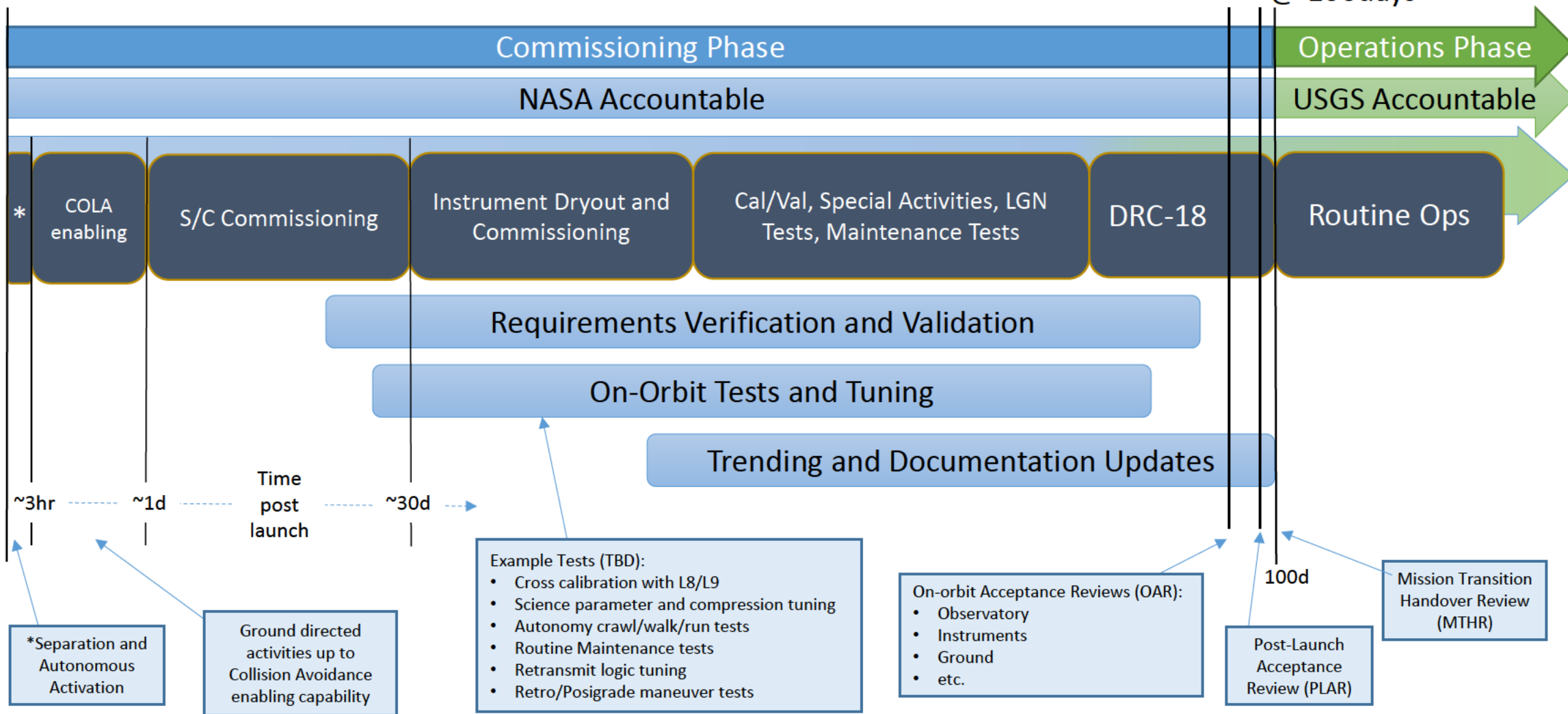
*Indicates sim use in preparation for SORT & MRT since the Run For Record will be executed against SC & Obs

- The L10 Mission baseline, as documented in Level 2 requirements, is 100 days for commissioning – this includes all activities up to handover and transition of operations to USGS
- The L10 Observatory will be injected into the operational orbit, as defined by the mission injection parameters
- Big picture phases (some activities can happen in parallel):
 - Launch, separation, autonomous deployments, autonomous transition to safe hold – approx. first 3 hours
 - Ground directed activities up to collision avoidance (COLA) propulsive maneuver enabling capability (Propulsion system checkout, Earth Pointing, etc.)
 - S/C commissioning, calibration, and checkout activities
 - LandIS instrument turn on, dryout, cooldown, calibration, and checkout activities
 - On-orbit verification, validation, calibration, tuning, and testing
 - Including maintenance demonstrations and LGN site testing
 - Complete nominal DRC-18 cycle
- Perform and finalize documentation updates, on-orbit performance comparison to pre-launch baseline, trending and end of life prediction updates, and anomaly summary/report closeout
- Mission Reviews/Milestones: On-Orbit Acceptance Reviews (OAR) – Observatory, Instruments, Ground, etc.; Post Launch Acceptance (PLAR); Mission Transition/Handover (MTHR)

Notional Commissioning Activities

Launch

Handover
@ 100days

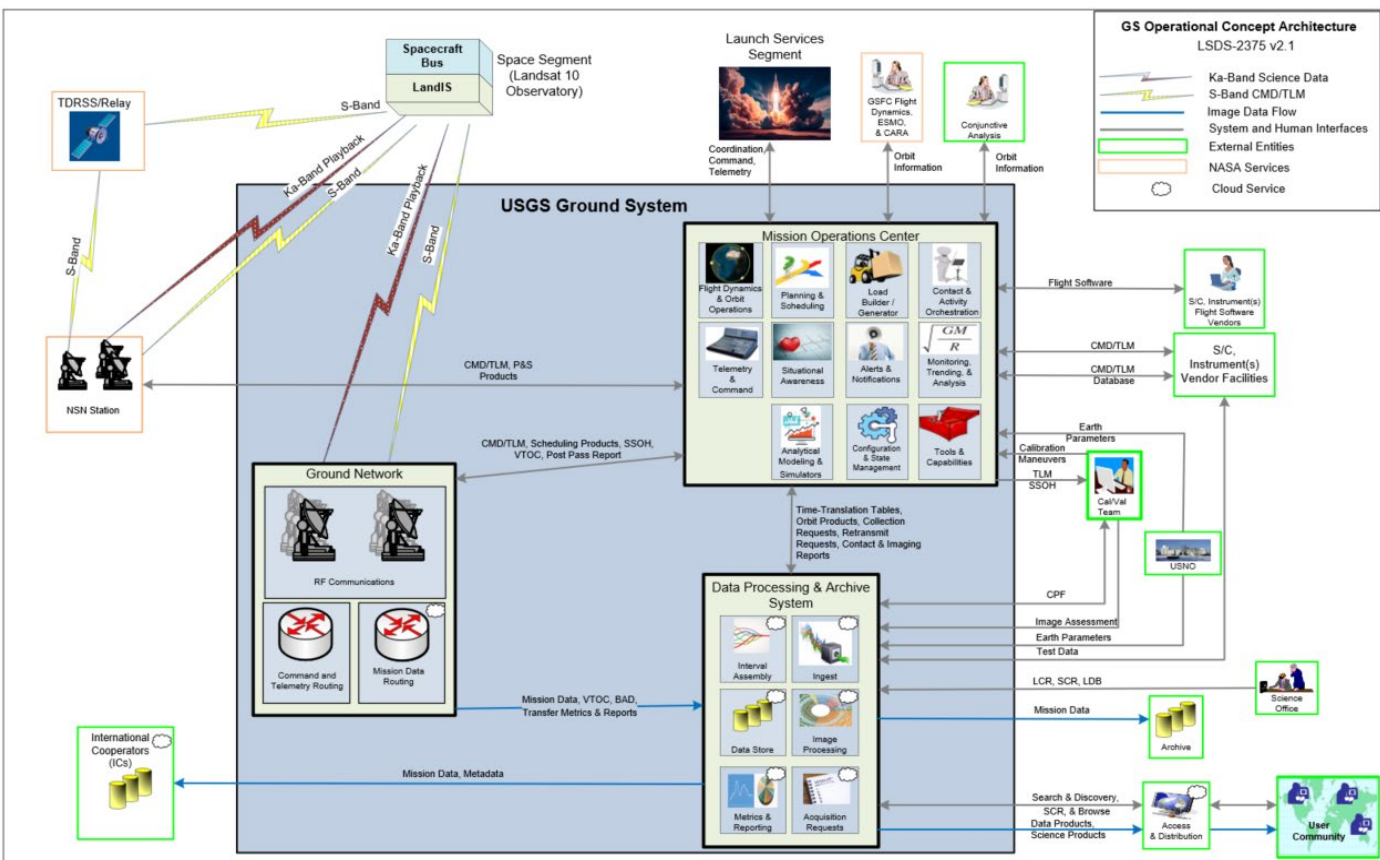


Mission Protection and Security Approach



- L10 is taking a proactive, comprehensive security approach, and will be following all relevant NASA and DOI/USGS guidelines.
- Please see the Landsat 10 Integrated Security Architecture Plan, L10-SYS-PLAN-0023, for an overview of the L10 Mission protection and security approach, including references to other key mission security documents.

Notional Depiction Only



• Mission Operations Center Functions

- **Telemetry and Command:** real time activities, and routine operations
- **Flight Dynamics / Orbit Operations:** orbit determination, ephemeris and event generation, and maneuver planning (orbit maintenance)
- **Planning and Scheduling:** contact, imaging, calibration, and all special activities to be executed by the Observatory
- **Analytical Modeling:** modeling and tools to verify current flight operations concepts and assess potential modifications
- **Simulators:** activity validation, anomaly support, training, product updates, etc
- **Trending and Analysis:** short-term, and long-term trending
- **Configuration & State Management:** manage observatory and ground
- **Contact & Activity Orchestration:** Plan, schedule, and manage all activities in support of lights out operations

• Ground Network (GN) Functions

- Provide the **RF interfaces** between the ground and Observatory
- **S-Band** interface for command and telemetry, and SSOH
- **Ka-Band** interface for the capture of mission data from the Observatory
- Mission **data transfer** to Landsat cloud

• Data Processing & Archive System Functions

- Generation of **special collection requests**, transmission to LMOC
- **Interval assembly and accounting** of mission data files received from the GN
- **Ingest** of mission data files to produce Level-0 intervals
- **Storing and archiving** the data and products (short-term and long-term storage)
- Level 1 and Level 2 **product generation**
- User **discovery and access**
- **International Cooperator** data and products exchange
- Ground system and science **metric** gathering and reporting

- The GN is comprised of the ground stations, networks, and all other data routing capabilities within the Ground System
- Key functions are to provide RF communications between the observatory and the ground for telemetry, command, and science data return
 - For telemetry, command, and stored state of health (SSOH) the GN provides the interface between the observatory and the MOC
 - For science data return, the GN provides the interface between the observatory and the DPAS
- S-Band RF uplink (encrypted) and downlink communications allow for MOC communications with the observatory
- Ka-Band RF receive communications allow for science data return
 - Science data return includes transfer of mission data files, including interleaved spacecraft and instrument ancillary data, and the volume table of contents (VTOC), for science data ingest, processing, and archival
 - During a Ka-Band mission data downlink event, the ground station monitors the downlink and provides a status of mission data files that were successfully received by the ground following the end of the contact
 - The Ground System uses this information to determine if retransmit commands are needed for specific science mission files
- Nominally 5-6 S-Band contacts (TBR) and 10-12 Ka-band contacts (TBD) per day for the observatory are expected
- The L10 GN is anticipated to be a global network employing a combination of polar and mid-latitude stations to optimize science data return and minimize operational costs
 - Alternative International Cooperator (IC) data flows based on downlink approach are in development
- The GN includes a cloud-based data routing capability for the transfer of science mission data from GN stations
 - Separation of mission data into priority-based virtual channels allows for expedited delivery of priority data
- Stations will cache Ka-Band mission data received for at least 7-days and S-Band data received for at least 7-days to support continuity of operations, contingencies, and retransfers of data

Mission Operations Center (MOC) Functions/Capabilities

Telemetry & Command (T&C)

Flight Dynamics/ Orbit Operations (FD/OO)

Planning & Scheduling (P&S)

Contact & Activity Orchestration (CAO)

Configuration Management & State Management

Trending & Analysis

Alerts & Notification

Load Builder/ Generator & Validation

Analytical Modeling & Simulators

Network and Infrastructure

Mission Operations Situational Awareness

Other Tools/ Capabilities

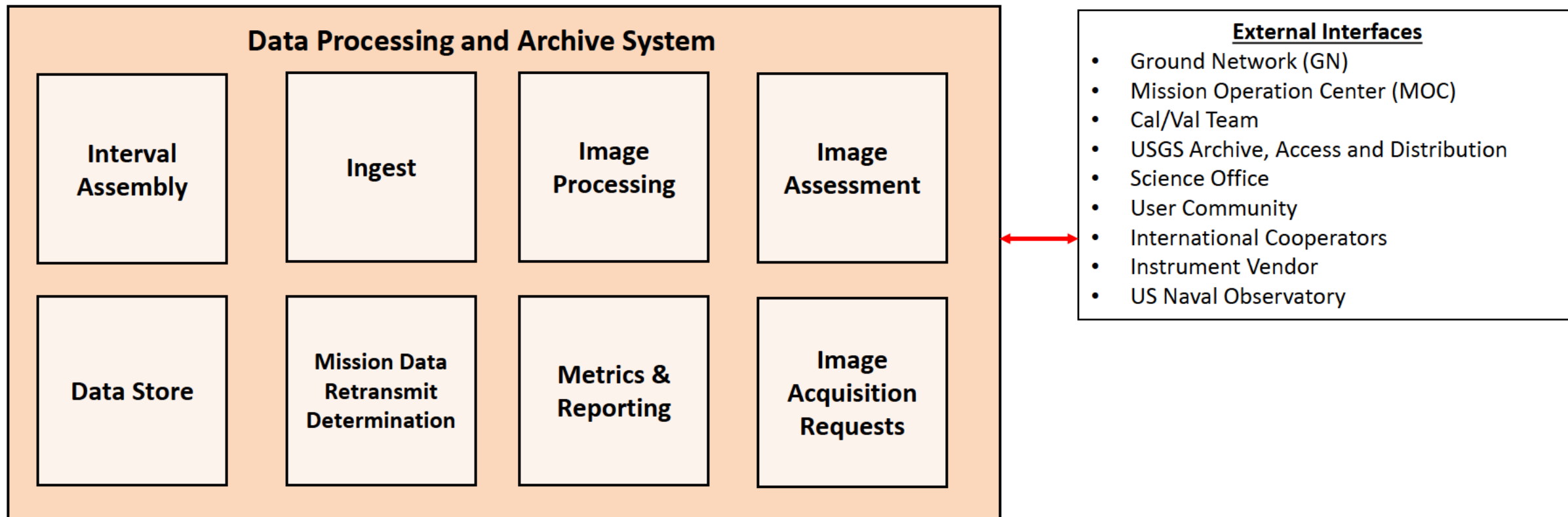
External Interfaces

- Ground Network (GN)
- Data Processing & Archive (DPAS)
- Cal/Val
- S/C Vendor
- Instrument Vendor
- Launch Services Segment
- NASA GSFC Services:
 - Flight Dynamics Facility (FDF)
 - Conjunction Analysis (CARA)
 - Near Space Network (NSN)
- US Space Command
- Status Dashboard
- Conjunction Analysis Services
- US Naval Observatory
- DoC/TraCSS

- The MOC includes all ground-based assets needed to operate the L10 mission, excluding the ground network, DPAS, LTAP generation, and retransmit decisions.
- MOC will support unattended (lights-out) operations for extended periods (days to weeks, TBR), and includes:
 - Autonomous contacts with ground stations for downlink of housekeeping and mission data
 - Autonomous contact changes/updates (due to station changes, new downlink needs, etc.) on ground and observatory
 - Autonomous notification of staff for any observatory or instrument issue or concern
 - Autonomous monitoring and reporting of MOC and GN (TBD) systems status, including notification of staff as required
- Mission operations includes a backup MOC (bMOC) – primary purpose of the bMOC is to maintain observatory health and safety, receive and process real-time telemetry, provide T&C capabilities, manage orbit and maneuver planning, and perform science data planning and scheduling functions to continue nominal operations.
- bMOC supports MOC resiliency in the event of critical failure
 - bMOC will assume operations of the mission within 24 hours (TBR) of loss of the primary MOC and is capable of providing MOC functionality for at least 30 days
- Pre-launch support for all factory and launch site observatory test events
- Test environment and connections to simulators for operational activity tests, product updates, and validation needs

- Mission Operations Management, Planning, Scheduling, Testing, and Orchestration
 - Contact Planning, Scheduling, and Deconfliction
 - Attitude and orbit determination capabilities and event propagation
 - Routine, special collection, and special activity request planning, constraint checking, deconfliction, verification, and loading
 - Routine orbit maintenance and risk mitigation maneuver planning, testing, constraint checking, and validation
 - Plan, schedule, and manage all routine contact activities, including routing and execution of retransmit requests from DPAS
 - Load building, validation, and uplink
 - Simulators for product testing, training, activity validation, anomaly investigations, or other engineering activities
 - Multi-satellite capable to accommodate L10 and any future observatory
- The system provides the engineering team the ability to confirm the observatory is imaging autonomously as defined in the LCR and SCR
 - Spacecraft provides the full, 18 day, up to date onboard WRS-3 scene (Path/Row) plan for validation on the ground
 - This imaging plan and deconfliction report is reconciled against the LCR/SCRs and the results are provided to the DPAS
- Autonomously review and validate the observatory provided 48 hour integrated onboard timeline (IOT) and notify engineering of any issues, errors, or unexpected results
- The MOC plans, schedules, and manages all flight operation resources and contact activities to support downlinking of mission data
 - Includes mass storage and priority data based reprioritization, changes/updates due to station outages/failures, etc.
 - Manage all mission data retransmission commanding
- The Analytical Model provides modeling and tools to verify current flight operations concepts and assess potential modifications, including assessing impacts of GN and imaging changes and outages

- Monitor and control the observatory and provide a full suite of analysis and tracking tools
 - Real-time and back-orbit (SSOH-based) monitoring of the housekeeping telemetry data, which includes traditional red and yellow “limit” monitoring, and configuration and state monitoring (for states, things not limited, or for tighter monitoring than limits)
 - Load building, testing, constraint checking, and real time commanding
 - Memory management, tracking, and comparison of ground “gold copy” (disk image, EEPROM, RAM, etc.) of onboard FSW, tables, and parameters
 - Life of mission event tracking, performance tracking, performance comparison, and reporting
 - Anomaly support, including remote access for engineering teams to support anomaly resolution
- Trending and analysis capability on all required timescales (real time, near-real time, periodic, long term, and life of mission) for spacecraft, instrument, and ground system telemetry and parameters
 - Automated plot/table/report generation for traditional daily, weekly, monthly, quarterly, yearly, and LOM
 - Custom plot/table/report generation on demand
 - Managing and reporting key performance parameters of the spacecraft, instruments, science data, and ground system segments for the life of the mission, including trends and margins
 - Managing key life limiting budgets (power, fuel, rotational components, etc.), including predictions of failure
- Configuration management – all flight and ground product and software version control, backup, and recovery
- State management – all flight and ground systems/equipment state and parameter configuration management



- The DPAS provides the functionality for data ingest, processing, archival, discovery, and data distribution or egress as required.
 - Assembly, ingest and processing includes assessing quality based on characterization and calibration updates
 - Produces collections of Landsat products enabling consistently characterized application by the user community
 - Primary interface for planning and managing science data acquisition requests
- Key DPAS functions:
 - Provide the capability to discover, receive, and fulfill user requests for L10 image collections and data products distribution
 - Provide the archive that holds and maintains all Landsat data and associated metadata
 - Performs interval assembly and accounting on mission data files received from the Ground Network, including decisions on retransmits of mission data files which were not successfully downlinked
 - Retransmit requests generated by DPAS will be sent to the MOC for command upload
 - Performs science data processing, including processing mission data to produce Level-0, Level-1, and Level-2 products
 - Provides an interface for users to discover and access data products
 - Provides functionality used by Government calibration and validation experts for all Landsat science data products
- Overall Ground System and Science Data metric gathering and reporting
 - Managing and reporting key performance parameters related to the ground system, science data, and the ground network, including (but not limited to) processing time, latency, completeness, quality, and file capture and loss rates
 - Collects metrics from calibration and validation on instrument characterization, performance and pointing accuracy

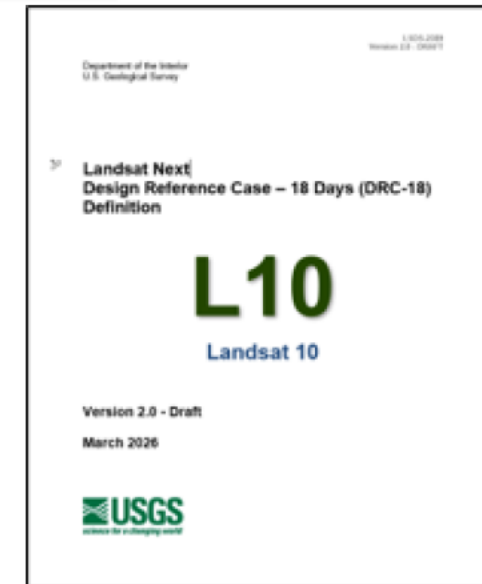
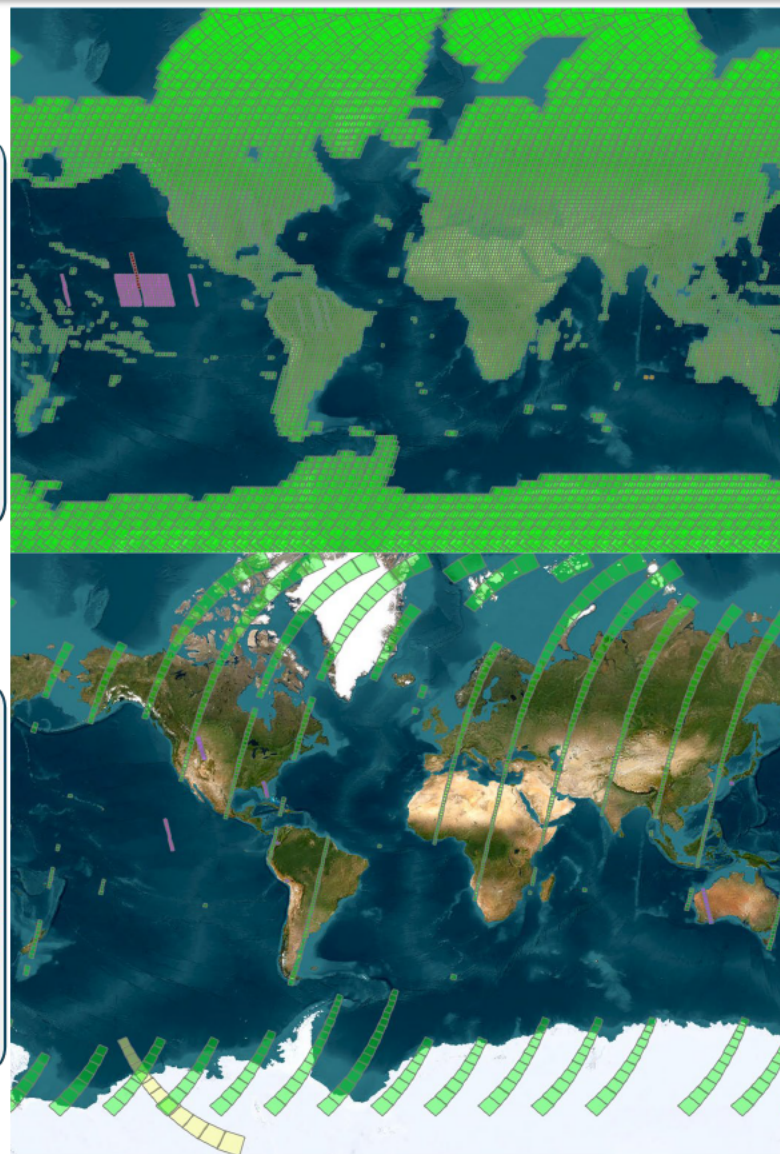
- Ground System supports Observatory buildup through additional capabilities external to the elements. These capabilities include:
 - Receiving test data from LandIS vendor to support early instrument characterization and inputs to software development
 - Ground System to Spacecraft Integration Racks (GSSIR)
 - Pseudo GN, DPAS and Cal/Val functionality deployed to spacecraft vendor facility
 - Collects and archives data from the Observatory during integration and test campaigns
 - Remotely accessible by USGS staff
- Landsat Science / Data Acquisition aids in the development of the LTAP Collection Request (LCR) for nominal sunlit Earth imaging and select night scenes
 - LCR is generally static
 - LCRs are completely replaced when required and not selectively updated onboard
 - LCR is augmented for special collection campaigns, emergency response, and calibration activities through the use of Special Collection Requests (SCRs)
 - SCRs have defined activities that replace the LCR activities (by Path/Row)
 - SCRs are frequently modified and/or replaced
 - SCRs have an effective start and end date, managed by the Ground System

18-day Design Reference Case (DRC-18)

- Developed to capture image acquisitions spanning the 18-day WRS-3 repeating ground track (referenced as DRC-18)
 - Assumes an “All-Land” acquisition model
 - Calibration activities represent the current Lands calibration approach
 - In total, ~800-1000 WRS-3 scenes are collected per day depending on time of year, amount of night collections, and various calibration activities
- Model and simulation of mission operations including the details of the DRC-18 image acquisition sequence and ground contacts developed
- DRC-18 will be used for observatory performance testing and validation, ensuring that Landsat 10 operations concept is incorporated into the test program (test as you intend to fly)
 - Performance Tests (e.g. mini-DRC in CPT), Day-In-The-Life scenarios, and for other testing and simulations, power modeling, etc.

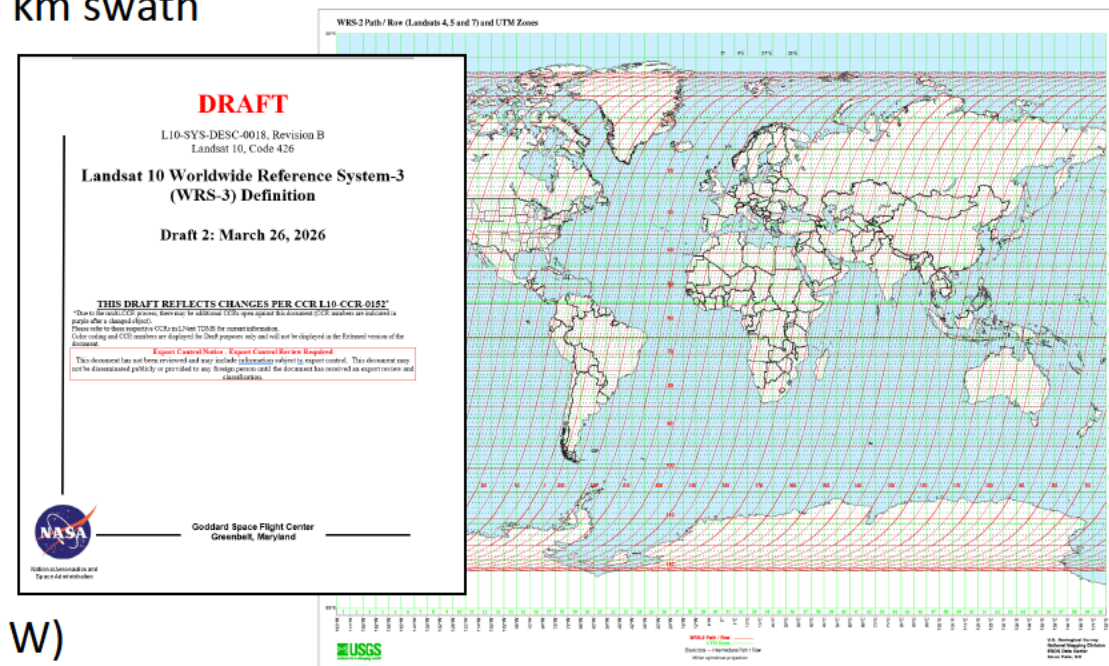
Legend
Collection Type 1 - Land Acquisition
Collection Type 2 - Lamp Calibration
Collection Type 3 - Solar Cal
Collection Type 4 - Reserved
Collection Type 5 - Lunar Cal
Collection Type 6 - Night Collection
Collection Type 7 - Off Nadir

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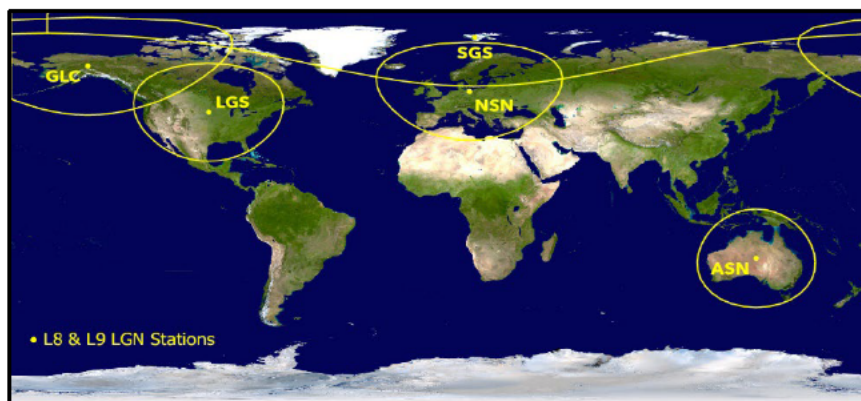


DRC Cycle Day	Scene Center Time	WRS-3 Path	WRS-3 Row	Collection Type	SunAngle
Day 1	Mar 11 2022 22:54:47.9	190	1		5.236715263
Day 1	Mar 11 2022 23:55:11.8	190	2		6.402812212
Day 1	Mar 11 2022 23:55:35.7	190	3		7.573601531
Day 1	Mar 11 2022 23:55:59.7	190	4		8.742835831
Day 1	Mar 11 2022 23:56:23.6	190	5		9.916264206
Day 1	Mar 11 2022 23:56:47.5	190	6		11.07562348
Day 1	Mar 11 2022 23:57:11.4	190	7		12.23866328
Day 1	Mar 11 2022 23:57:35.4	190	8		13.39999101
Day 1	Mar 11 2022 23:57:59.3	190	9		14.55662944
Day 1	Mar 11 2022 23:58:23.2	190	10		15.71098025
Day 1	Mar 11 2022 23:58:47.1	190	11		16.86189723
Day 1	Mar 11 2022 23:59:11.1	190	12		18.00887733
Day 1	Mar 11 2022 23:59:35.0	190	13		19.15178475
Day 1	Mar 11 2022 23:59:58.9	190	14		20.2901458
Day 1	Mar 12 2022 00:00:22.8	190	15		21.02448808
Day 1	Mar 12 2022 00:00:46.7	190	16		22.15234602
Day 1	Mar 12 2022 00:01:10.7	190	17		23.27512513
Day 1	Mar 12 2022 00:01:34.6	190	18		24.39171152
Day 1	Mar 12 2022 00:01:58.5	190	19		25.50818186
Day 1	Mar 12 2022 00:02:22.4	190	20		26.60595628
Day 1	Mar 12 2022 00:02:46.4	190	21		27.70991106
Day 1	Mar 12 2022 00:03:10.3	190	22		28.78877141
Day 1	Mar 12 2022 00:03:34.2	190	23		29.86011977
Day 1	Mar 12 2022 00:03:58.1	190	24		30.93332989
Day 1	Mar 12 2022 00:04:22.1	190	25		31.99879029
Day 1	Mar 12 2022 00:04:46.0	190	26		33.04881767
Day 1	Mar 12 2022 00:05:09.9	190	27		34.08769333
Day 1	Mar 12 2022 00:05:33.8	190	28		35.11465114
Day 1	Mar 12 2022 00:05:57.8	190	29		36.12888713
Day 1	Mar 12 2022 00:06:21.7	190	30		37.12553925
Day 1	Mar 12 2022 00:06:45.6	190	31		38.11569703
Day 1	Mar 12 2022 00:07:09.5	190	32		39.08839251
Day 1	Mar 12 2022 00:07:33.4	190	33		40.04069171
Day 1	Mar 12 2022 00:07:57.4	190	34		40.9723841
Day 1	Mar 12 2022 00:08:21.3	190	35		41.88615779

- A new global grid system called WRS-3 developed to correspond to the 653 km repeating ground track
 - Conceptually identical to the WRS-2 used for previous Landsat missions
- The 18-day cycle consists of 265 orbits; whereby the WRS-3 has 265 paths (compared to 233 for WRS-2)
- The 151-km equatorial separation between adjacent orbital paths results in 164 km instrument minimum swath width after adding 10 km for ground track error and 3 km for margin
 - Same approach used for WRS-2, with 172 km separation & 185 km swath
- The along track scene length is 168 km which includes 1.5% in-track overlap with adjacent scenes
 - This results in the WRS-3 having 248 rows (drawn parallel to latitude) which is the same number as the WRS-2 system
- Each intersection of a path and row identifies the center point of a unique Landsat scene
 - Centers spaced 162km (23.66s increments of spacecraft time)
 - The rows are indexed so that the scene center at the orbital descending node is Row 60
 - The longitude of Path 001, Row 60 is set to -65.2 degrees (65.2 W)



- Utilization of key international ground station partners in Australia and Germany to downlink and transfer data to the archive at USGS/EROS in Sioux Falls, SD



Alice Springs, Australia



Neustrelitz, Germany

Illustration Source and Photo Credit: USGS

State of the Landsat Collection 2 consolidated global data archive

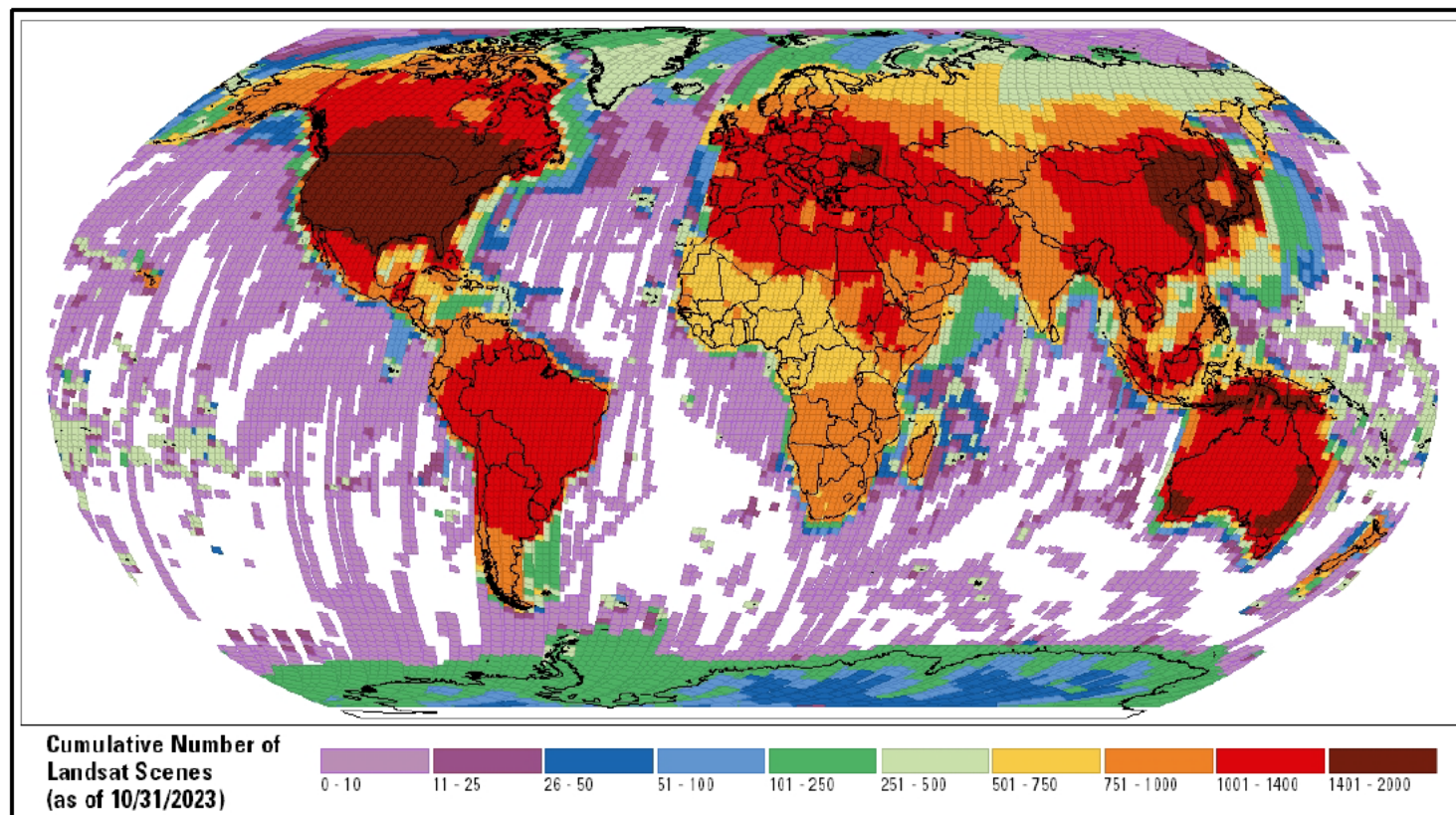
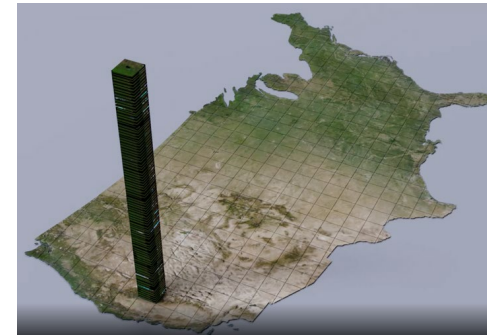
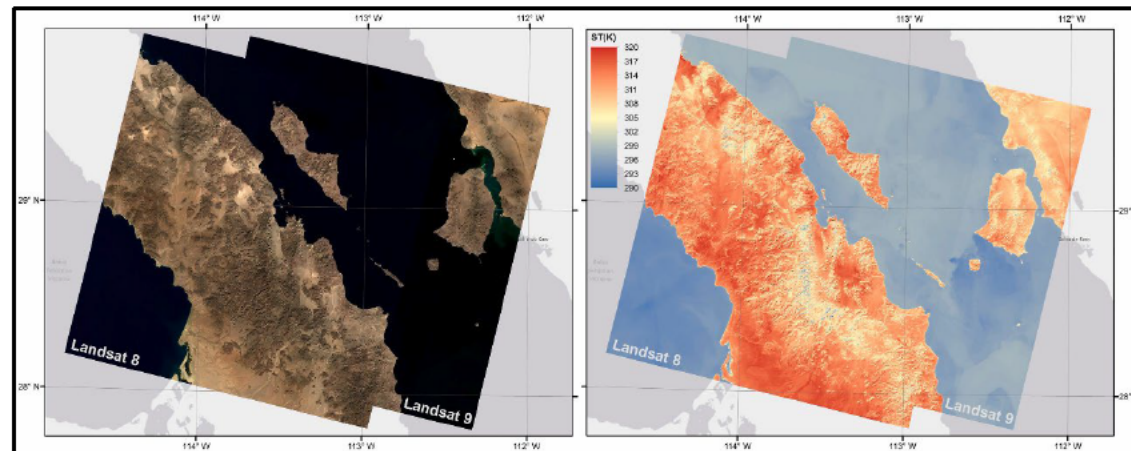


Illustration Source: USGS

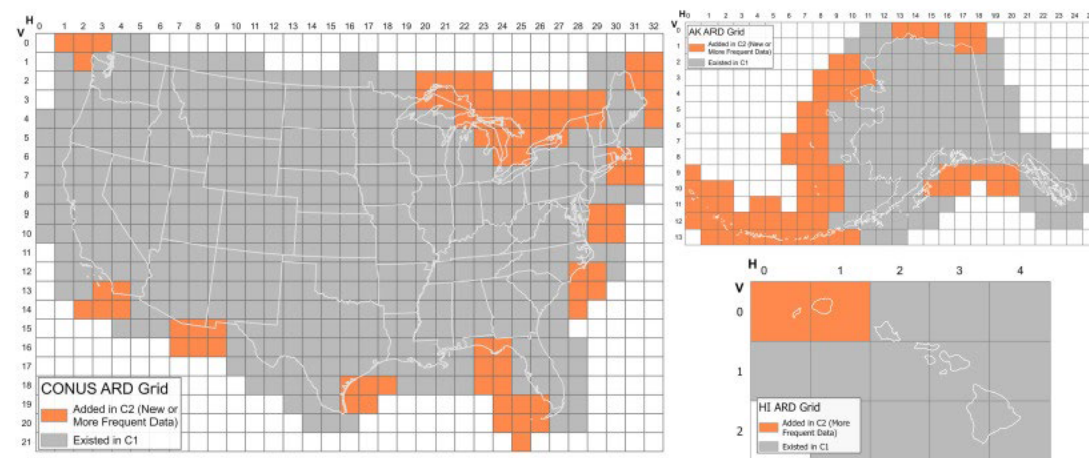
- L10 science mission products will include the following in either scene or tiled formats for VSWIR and TIR measurements:
 - Level-1 Top of Atmosphere (TOA) calibrated digital counts, view angles, and metadata
 - Level-2 Surface Reflectance, Aquatic Reflectance, Surface Temperature, and Surface Emissivity geophysical quantities, view angles, and metadata
- L10 science algorithms (i.e., atmospheric compensation) for product generation will require evolution and advances:
 - Leverage new atmospheric and surface spectral measures on-board (e.g., aerosol, water vapor, emissivity)
 - VSWIR algorithms will combine heritage and emerging techniques
 - Will include a Nadir BRDF-Adjusted Reflectance (NBAR) Correction
 - TIR algorithms will combine thermal-emissivity separation (TES) and generalized dual channel techniques
- L10 science mission products will be processed into the existing Landsat Collection at time of operations



- The Landsat consolidated global archive has been processed twice by the USGS, Collection 1 in 2016, and Collection 2 in 2020
- The Landsat Collection 2 archive is a 52+ year record with >12 million images and > 10 petabytes
- See Crawford et al. (2023). The 50-year Landsat Collection 2 archive. *Science of Remote Sensing*. 8, 100103 
- USGS is currently modernizing the Landsat Data Processing, Archive, access System (DPAS) to be fully cloud service enabled
- USGS is currently formulating Landsat Collection 3 for another archive reprocessing in late 2020s to prepare for Landsat 10's data archive integration



Landsat 8/9 Collection 2 surface reflectance and temperature images during November 2021 Landsat 9 ascent underfly campaign, Baja California, Mexico



Landsat tiled analysis ready data (ARD) concept, Crawford et al. (2023)

Landsat Open Access Data, Users, and Economic Value

- Landsat archive data products have been open access and online since 2008
- The authoritative Landsat Collection 2 archive is accessible from USGS portals (illustration below)
- Data volume disseminated, user-base, and economic values continues to accelerate

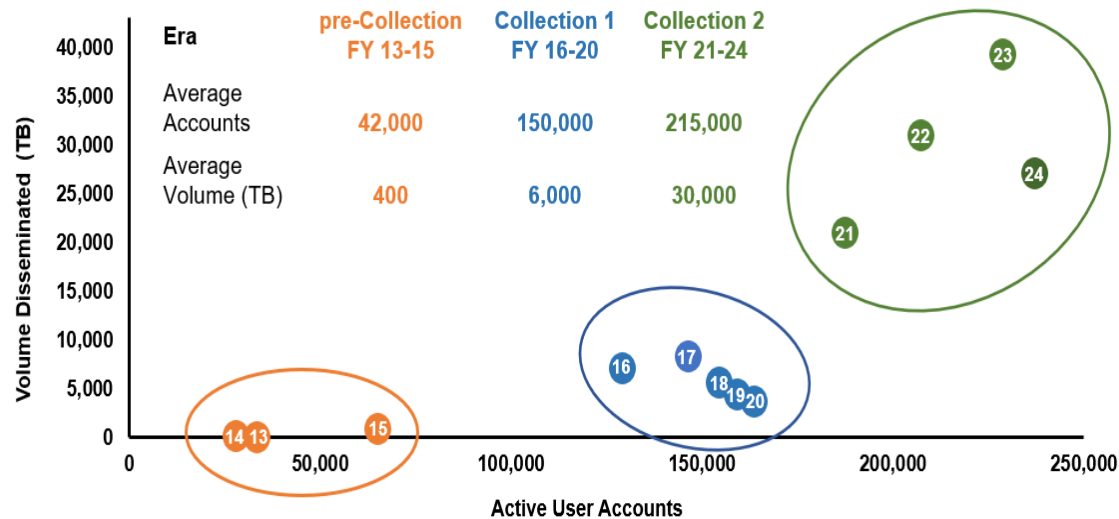


Illustration Source: USGS

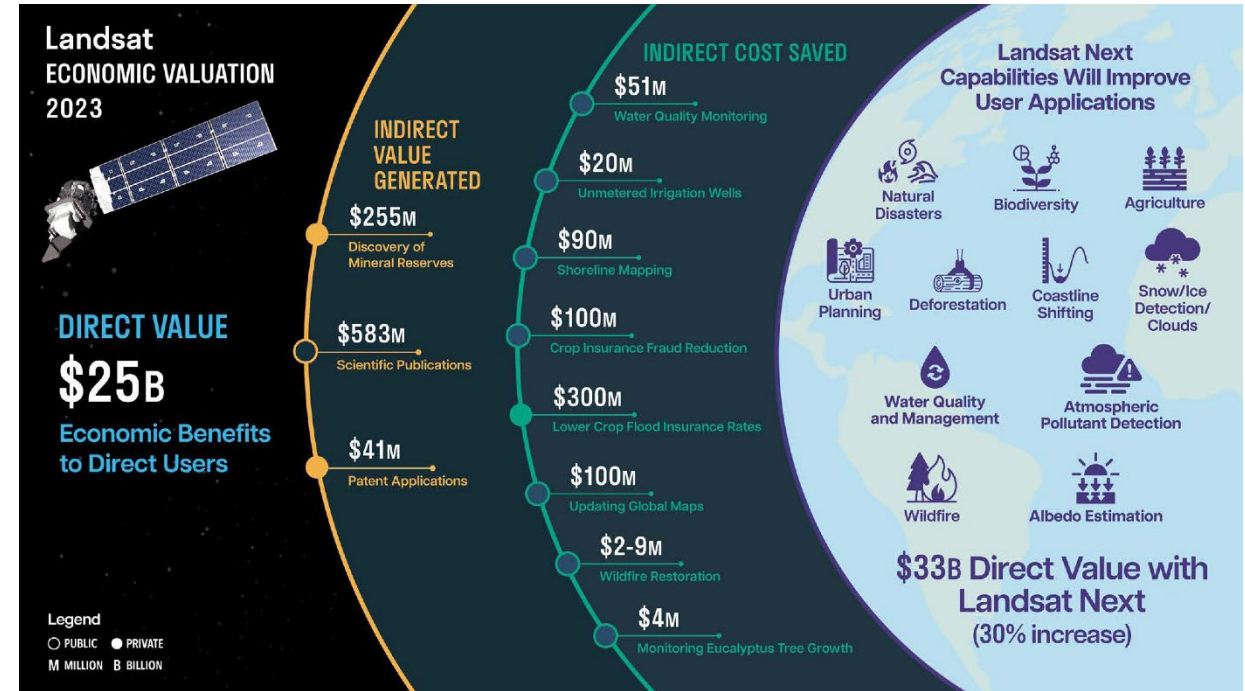
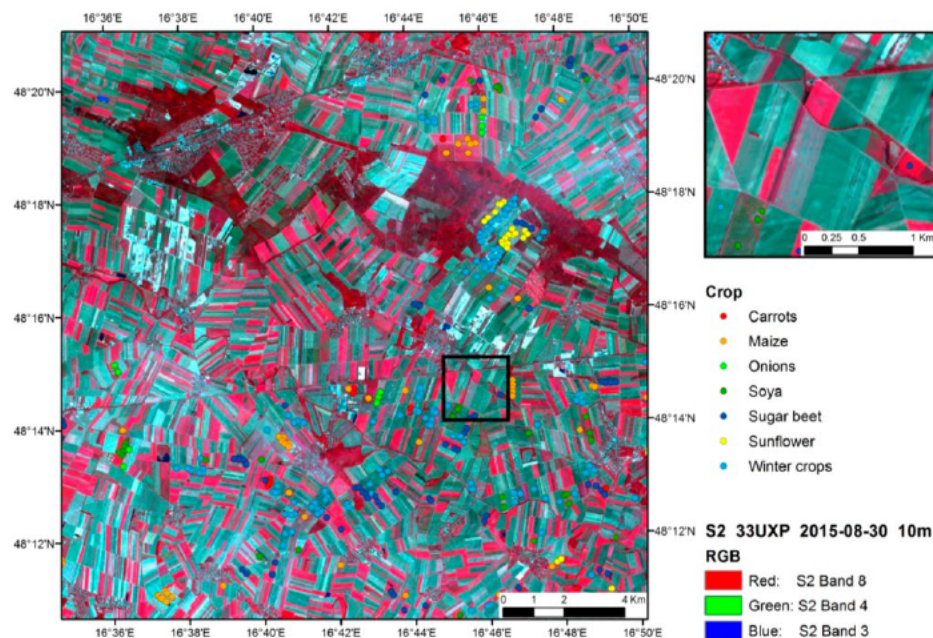


Illustration Source: USGS contracted report on *Economic Valuation of Landsat and Landsat 10, 2023*

- **L9 → Landsat 10:**
 - 30 → **10m** RGB, NIR, and SWIR1 (1610 nm)
 - 30 → **20m** other surface reflective VSWIR bands
 - 100 → **60m** atmospheric VSWIR and TIR bands
- **E.g., Agriculture:** Support field-level crop mapping, yield estimates in small/irregular fields, arable land and water-use estimates, urban areas, surface water area, water quality in rivers/lakes/coastal areas, vegetation type and conditions, surface temperature and composition.



Immitzer et al. 2016

Lesiv et al. (2018) stated that moderate resolution data such as Landsat at 30m missed smaller farms that occupy about 40% of global croplands

Inter- and intra-field heterogeneity observed at the S2 spatial resolution of 10m.



L10 Science Requirements: Spectral & Spatial Table

	Band Name	Ground Sample Distance (m)	Center wavelength (nm)	Band width (nm)	Rationale
1	Violet	60	412	20	Improved aerosol retrieval; CDOM from inland/coastal water
2	Coastal Aerosol	20	443	20	Landsat
3	Blue	10	490	65	Landsat
4	Green	10	560	35	Landsat
5	Yellow	20	600	30	Leaf chlorosis, vegetation stress and mapping
6	Orange	20	620	20	Phycocyanin detection for Harmful Algal Blooms
7	Red 1	20	650	20	Phycocyanin, chlorophyll
8	Red 2	10	665	30	Landsat
9	Red Edge 1	20	705	15	LAI, Chlorophyll, plant stress (S2)
10	Red Edge 2	20	740	15	LAI, Chlorophyll, plant stress(S2)
11	NIR_Broad	10	842	115	10m NDVI (S2)
12	NIR1	20	865	20	Continuity (note-S2 narrower than L8)
13	Water vapor	60	945	20	Improved atmospheric correction for LST, SR (S2)
14	Liquid Water	20	985	20	Liquid water, water surface state
15	Snow/Ice 1	20	1035	20	Snow grain size for water resources
16	Snow/Ice 2	20	1090	20	Ice absorption, snow grain size
17	Cirrus	60	1375	30	Landsat
18	SWIR 1	10	1610	90	Landsat
19	SWIR 2a	20	2038	25	Subdivided for cellulose/crop residue measurement (Landsat)
20	SWIR 2b	20	2108	40	Subdivided for cellulose/crop residue measurement (Landsat)
21	SWIR 2c	20	2211	40	Subdivided for cellulose/crop residue measurement (Landsat)
22	TIR 1	60	8300	250	Mineral and surface composition mapping (ASTER)
23	TIR 2	60	8600	350	Emissivity separation, volcanos (SO2) (MODIS/ASTER)
24	TIR 3	60	9100	350	Mineral and surface composition mapping (ASTER)
25	TIR 4	60	11300	550	Surface temperature (Landsat), carbonates
26	TIR 5	60	12000	550	Surface temperature, snow grain size (Landsat)

Landsat 10 - Spectral Band Comparison Tool

NASA tool that compares spectral bands across Landsat **and other Earth-observing instruments** by wavelength and GSD.
<https://landsat.gsfc.nasa.gov/spectral-band-comparison-tool/>

